# Collinear DFT Tests of Materials Project Magnetic-Ordering Labels Among Dzyaloshinskii–Moriya Screening Candidates

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## Abstract

The Materials Project (MP) is the usual first filter in searches for Dzyaloshinskii–Moriya-interaction (DMI) hosts, yet its workflow starts every calculation from a ferromagnetic (FM) guess, and for most entries no antiferromagnetic (AFM) alternative has been computed. We test MP's stored net-moment labels with collinear density-functional theory (DFT) in the population that DMI screening selects: noncentrosymmetric compounds containing a heavy spin–orbit element. Converging symmetry-distinct collinear orderings for 26 candidates, we find that 17 lose their net moment to an AFM ground state (margins 0.19–24.68 meV atom$^{-1}$), 3 keep a net moment as ferrimagnets, 4 are consistent with their stored net-moment label, and 2 are degenerate. Seven of the 17 losses are in compounds whose MP cell holds a single Fe site, where no AFM order could have been computed; the other 10 are in cells that could represent the AFM state but were never started from it. The batch is dominated by Fe oxides, for which superexchange favours antiparallel alignment, so these numbers quantify how unreliable net-moment labels are within this screening family rather than across MP. For two compounds whose Fe sublattice remains FM, $Li_4Fe_2TeWO_{12}$ and $Tb_6FeBi_2$ (Tb $4f$ in the core), we extract exchange from DFT, and DMI for the first ($|\mathbf{D}|/J \approx 5.6\%$; it vanishes by symmetry in the second); classical atomistic spin dynamics with the unmodified exchange places the ordering temperatures of these models near 100 K and 140 K. MP ordering labels should be treated as a screening prior, not ground truth, for DMI-material discovery.


## INTRODUCTION

Materials that host a stable Dzyaloshinskii–Moriya interaction (DMI) [1, 2] are of active interest for chiral spin-texture and skyrmion-based non-volatile memory, because the antisymmetric exchange term the DMI contributes can stabilize spin spirals and skyrmion lattices at zero field [4–6]. Skyrmion lattices of this kind have been observed in bulk chiral magnets [7, 8], at atomic scale in ultrathin films [9], written and deleted individually with a spin-polarized current [10], and stabilized at room temperature in multilayer stacks [11]; the materials landscape and the device prospects that motivate this search are


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reviewed in refs. [12, 13]. Searching for such materials computationally typically begins with a curated high-throughput database – most commonly the Materials Project (MP) [14] – used to filter tens of thousands of candidates down to a tractable shortlist before committing expensive first-principles or simulation resources. That filtering step depends on the database's recorded magnetic ordering (ferromagnetic, antiferromagnetic, ferrimagnetic, or non-magnetic) being correct, or at least an unbiased estimate.

Automated high-throughput DFT workflows are prone to a specific failure mode here: relaxing a structure from a single initial magnetic-moment guess – almost always ferromagnetic – converges to whichever local minimum that guess is closest to, without necessarily finding the true magnetic ground state [15]. Materials Project's own magnetic-ordering methodology paper is explicit that a full energy-mapping analysis, enumerating symmetry-distinct antiferromagnetic configurations and comparing converged energies, is required to determine the ground state reliably [15], whereas entries generated at high throughput are relaxed from a single ferromagnetic initial guess.

Recent independent evidence, obtained through a very different route, points at the same problem. Fahmy [16] trained propagation-vector and magnetic-order classifiers on the MAGNDATA database of experimentally solved magnetic structures [17, 18] and applied them to the Materials Project, finding a systematic disagreement consistent with a ferromagnetic bias affecting more than 7,800 MP-labeled compounds. That is a large-$N$, purely statistical detection: the classifier disagrees with the label, but no first-principles calculation is performed on any individual compound to confirm which side is right.

Here we ask the complementary question at higher fidelity and lower $N$: within the structural family that DMI-candidate screening pipelines actually select – noncentrosymmetric space groups, a net magnetic moment, and a heavy spin-orbit-coupling (SOC) element – how often is MP's stored FM label the true DFT ground state? We build a seven-source harmonized materials database and a structure-aware magnetic-ordering classifier, use them to filter for DMI-compatible candidates, and run the collinear energy-mapping protocol MP itself prescribes [15] on 26 shortlisted candidates. The result is a per-compound DFT verdict rather than a statistical disagreement: of the 26 net-moment labels (24 FM, 2 ferrimagnetic), 17 lose their net moment to an AFM ground state and 3 more change from FM to ferrimagnetic. That FM-initialized workflows miss AFM ground states is documented by MP itself [15]; what we add is its size in the population that DMI screening selects, the sepa-

ration of the cases MP's cell could not represent from those it simply did not try, and a DFT-to-spin-model chain for the candidates that survive. Twenty-four of the 26 candidates, and all but one of those that keep a net moment, are hypothetical structures generated by MP's substitution workflows, which is itself a property of what DMI screening selects. Four candidates keep a ferromagnetic Fe sublattice, and they are three compounds, because the two $Tb_6FeBi_2$ entries (mp-22972 and mp-1079285) relax to the same structure. We carry two of these compounds, $Li_4Fe_2TeWO_{12}$ (mp-754090) and $Tb_6FeBi_2$ (modelled on mp-1079285), from DFT exchange constants, and for the first the DMI, to finite-temperature classical atomistic spin dynamics with no fitting or rescaling, and fix the ordering temperature of each model by finite-size scaling; for the third, $FeP_6(WO_8)_3$ (mp-850225), we resolve its two exchange shells.

## RESULTS

### Screening 64,635 records to a DMI-compatible shortlist

Seven public sources – Materials Project [14], JARVIS-DFT [19], AFLOW [20], MAGNDATA [17, 18], 2DMatPedia [21], OQMD [22, 23] and NOMAD [24] – were merged into one schema of 64,635 records (Methods). Where no ground-truth ordering exists, it is predicted by a LightGBM [25] classifier on composition, global-structure and local-magnetic-geometry features, which reaches 67.4% accuracy (68.0% balanced accuracy) on a stratified held-out set of 7,193 records (Supplementary Fig. 1). Its dominant confusion is between FM and ferrimagnetic (FiM), the two net-moment classes that the screening treats identically. Graph neural networks trained and tested on the same split – a crystal graph convolutional network learned from scratch and LightGBM probes of the frozen MACE-MP-0 foundation model – reach 63.3–74.6% accuracy (65.9–71.7% balanced; Supplementary Table 1, Methods). The best, which appends the MACE-MP-0 embedding to the screening features, gains 7 points of accuracy, but in agreement with Materials Project's own labels: on the experimentally labelled MAGNDATA subset every model lies between 49 and 56% balanced accuracy, within the sensitivity of that small set. Retrained with the 26 DFT-validated candidates of this work held out, the models call 6–12 of the 20 whose DFT ground state is not FM non-ferromagnetic, 1.5–2.7 times their rate on Materials Project ferromagnets. The screening

below uses the LightGBM classifier throughout.

The database is then filtered for candidates plausibly compatible with a DMI (Fig. 1a): a net magnetic moment (FM or FiM), a noncentrosymmetric space group, at least one heavy-SOC element, and thermodynamic plausibility. The symmetry filter is a necessary condition only. Moriya's rules forbid the DMI on a bond whose midpoint is an inversion centre [2], so centrosymmetric crystals can still carry bond DMI (and weak ferromagnetism); what a bulk inversion centre removes is the net Lifshitz invariant that stabilizes homochiral spirals and skyrmions. Conversely, the noncentrosymmetric point groups $\bar{4}3m$, $\bar{6}$ and $\bar{6}m2$ also have no Lifshitz invariant [3, 4]; 901 of the 1,864 records that pass the heavy-SOC filter (797 of them in $F\bar{4}3m$) and 4 of the 26 DFT candidates lie in these classes, and we keep them because bond-level DMI, and interfacial or strain-induced chirality, remain possible. The heavy-SOC list (Ta, W, Re, Os, Ir, Pt, Au, Hg, Bi) is a choice: it follows the $5d$ and $6p$ species of the Fert–Levy and Moriya mechanisms [2, 26–28] and omits other strong-SOC elements such as Pb, Tl, Pd, Te, I and the lanthanides, which enter the shortlist only alongside a listed element. Of 64,635 records, 47,614 carry a net-moment ordering, 19,280 of those are noncentrosymmetric, and 1,864 also contain a heavy-SOC element (Materials Project 768, JARVIS 537, AFLOW 392, 2DMatPedia 77, NOMAD 53, OQMD 35, MAGNDATA 2). Dropping every record with a known hull distance above 100 meV atom$^{-1}$ (sources without a hull field are retained) leaves 1,195 (Fig. 1b,c). The heavy-SOC filter shifts the population sharply towards AFLOW, and the hull filter towards the sources with no hull energy to test. After a structure-backfill and machine-learned-potential consistency check of the top of each source's ranking (Methods), the resulting shortlist is the pool from which the DFT batch of 26 Materials Project candidates is drawn, every one carrying a net-moment ordering computed by MP (24 FM, 2 FiM), which our harmonized schema records as a ground-truth label with confidence 1.0. All 26 survive the hard hull filter, and they spread from rank 1 to rank 663 of the hard-filtered ranking, so the batch is a sample of the funnel's DMI-compatible population rather than of its most stable members (Supplementary Fig. 2).

**DFT removes the net moment of 17 of 26 candidates**

For each candidate we enumerate FM and symmetry-distinct collinear AFM configurations and relax them with MP's GGA+$U$ (generalized gradient approximation plus Hubbard

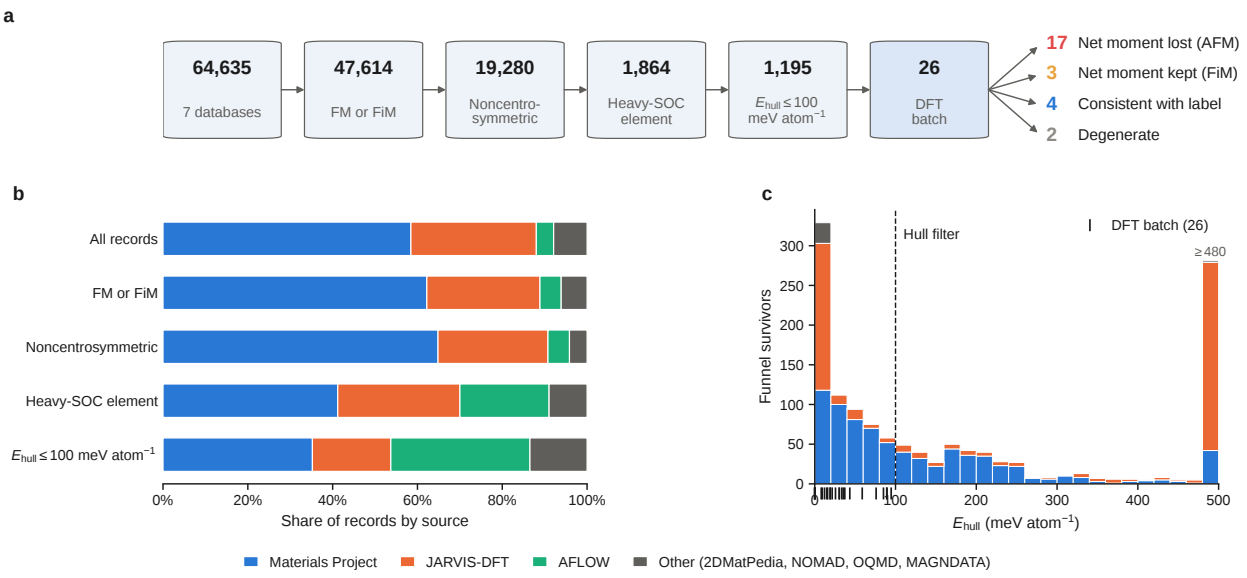



Fig. 1. **Screening a seven-source database for DMI-compatible ferromagnets.** **a**, The screening funnel: net-moment ordering (ground-truth label where present, otherwise the classifier's prediction), noncentrosymmetric space group, at least one heavy-SOC element, and a known hull distance no more than 100 meV atom$^{-1}$ above the hull (records without one are retained), ending in the 26-candidate Materials Project batch tested with DFT and its verdicts. **b**, Share of the records at each stage by source database (legend shared with **c**); "Other" pools 2DMatPedia, NOMAD, OQMD and MAGNDATA. **c**, Hull distance of the 1,340 heavy-SOC survivors that carry one (Materials Project and OQMD as stored, JARVIS-DFT recomputed from its own formation energies; Methods), in 20 meV atom$^{-1}$ bins with values $\geq$ 480 meV atom$^{-1}$gathered in the last bin; dashed line, the hull filter; ticks below the axis, the 26 DFT candidates.

$U$) recipe, audit every calculation for self-consistency and for the intended per-ion spin state, and re-decide every small margin with a same-cell ("tight") FM/AFM pair (Methods). A configuration is taken as the ground state when it lies more than 0.10 meV atom$^{-1}$ below every other audited configuration, and a candidate is called degenerate when its two lowest orderings differ by less than this; a candidate is resolved only when every staged configuration has a valid energy, or when an audited non-FM configuration already lies below FM, so that the missing ones could only deepen a non-FM verdict. The window is not set by numerical noise: a 1.5-fold denser $k$-mesh and a 650 eV cutoff move the two smallest tight margins, both insulating, by at most 0.001 meV atom$^{-1}$; for metals the $k$-mesh matters

more (Methods). It is a conservative allowance for the protocol and functional choices that a margin of this size cannot outlast.

We sort the outcomes by what they mean for a screening that keeps net-moment compounds (Fig. 2a, Table 1). *Net moment lost*: 17 candidates, all labelled FM, have a fully compensated AFM ground state (margins 0.19–24.68 meV atom$^{-1}$, 1.9–230 meV per Fe). *Net moment kept*: three FM-labelled candidates have a ferrimagnetic ground state that keeps a net moment – $Li_2Fe_3WO_8$ (mp-775194, 8 $\mu_B$ per cell), $Fe_3Co_2P_6WO_{24}$ (mp-769471, whose lowest cell is AFM between Fe sites but leaves 4.0 $\mu_B$ uncompensated) and $FeP_6(WO_8)_3$ (mp-850225), whose Fe sublattice stays FM by 1.81 meV atom$^{-1}$ but whose $W^{5+}$ moments align antiparallel to it, 2.0 $\mu_B$ per formula unit against the 8 $\mu_B$ MP stores (at $U_W = 0$; MP's $U_W = 6.2$ eV did not converge, Methods); for a net-moment screening these are not errors. *Consistent with the label*: 4 candidates, the ferromagnets $Li_4Fe_2TeWO_{12}$ (mp-754090) and $Tb_6FeBi_2$ (mp-1079285), 2.23 and 1.24 meV atom$^{-1}$ below the next configuration, and two FiM-labelled entries: $CrFe(BiO_3)_2$ (mp-551086), whose lowest state is a ferrimagnetic Fe/Cr cell 0.80 meV atom$^{-1}$ below FM, and the other $Tb_6FeBi_2$ entry (mp-22972), whose Fe sublattice is FM by 1.16 meV atom$^{-1}$ at a converged $k$-mesh with small induced Tb 5$d$ moments (0.05 $\mu_B$) antiparallel to Fe (Methods). *Degenerate*: 2, $CaLaTaFeO_6$ (AFM 0.10 meV atom$^{-1}$ below FM) and $Ba_4Ta_{10}FeO_{30}$ (0.00 meV atom$^{-1}$), both from tight pairs, which are evidence for neither ordering. No candidate remains unresolved: the one configuration never completed (the $1 \times 1 \times 2$ stacking of mp-682554) belongs to a candidate that another audited AFM cell already places 0.32 meV atom$^{-1}$ below FM at a converged $k$-mesh. All verdicts are among collinear orderings; a noncollinear or incommensurate ground state, which is exactly what a DMI host may adopt, is outside the enumeration, so "consistent with the label" means lowest among the collinear states tested.

How a label can be wrong depends on MP's cell. Eleven candidates have a single Fe site in the cell MP computed (counting the Fe-only sublattice; the Ti of $TiFeBiPbO_6$ is $d^0$), and in such a cell no Fe–Fe AFM order can be represented: the FM label then means only that the compound is magnetic. Seven of these 11 lose their net moment, two are consistent with their label (the two $Tb_6FeBi_2$ entries), one keeps a net moment as a Fe–W ferrimagnet (mp-850225) and one is degenerate. The other 15 have two or more magnetic sites in MP's cell, where an AFM state was representable but was not reached from the FM start: 10 lose their net moment, 2 keep it as ferrimagnets, 2 are consistent with their label

and 1 is degenerate. The loss rate is therefore similar in both groups, 7/11 (95% Wilson interval 0.35–0.85) and 10/15 (0.42–0.85), and 17/26 = 0.65 (0.46–0.81) overall. The 26 are not independent chemistries: three (A,A′)$FeBiO_6$ double perovskites with identical margins, two $Ta_2FeO_6$, two $TaFeO_4$ and two $Tb_6FeBi_2$ entries leave 21 distinct compositions, of which 13 lose their net moment (0.41–0.79).

The chemistry of the batch makes this outcome expected rather than surprising. Twenty-three of the 26 are oxides, nearly all $Fe^{2+}$ or $Fe^{3+}$ oxides in which the magnetic cations are bridged by oxygen – the geometry for which Goodenough–Kanamori–Anderson superexchange between $d^5$ or $d^6$ shells favours antiparallel alignment [29–31]. A workflow that initializes every relaxation ferromagnetically starts each of these compounds from the chemically less likely state, and MP's own magnetic-ordering study [15] anticipates the result. What the batch measures is how large the effect is inside the population that DMI screening selects. The smallest losses need individual comment. $FeCu_4Re_{14}(Mo_3S_{20})_2$ (mp-682554) has one Fe per 65-atom formula unit; its 0.32 meV atom$^{-1}$ preference (21 meV per Fe, unchanged on a $k$-mesh 3-fold denser along each axis) couples Fe sites at least 6.8 Å apart through a metallic Re–Mo–S cluster framework, so it is a long-range, presumably RKKY-like, coupling that a collinear GGA+$U$ calculation can describe only qualitatively. The tight pairs of $Zn_2FeWO_6$ and the two $Ta_2FeO_6$ entries ($-0.19$ to $-0.41$ meV atom$^{-1}$) are within a factor of a few of the 0.10 meV atom$^{-1}$ window and are the verdicts most exposed to protocol noise (Methods). That exposure is real for metals: the tight pair of mp-22972, at MP's $k$-point density (a single $k$-point along its 14.5 Å axis), placed an AFM stripe 0.46 meV atom$^{-1}$ below FM, and a uniform ∼60 Å $k$-length reverses it to 1.16 meV atom$^{-1}$ above FM (Methods).

The tight pairs changed four verdicts, in both directions. $Zn_2FeWO_6$ (mp-1047414), degenerate at 0.10 meV atom$^{-1}$ under the MP recipe, is AFM by 0.17–0.19 meV atom$^{-1}$ in both of its same-cell pairs; $Ba_4Ta_{10}FeO_{30}$ (mp-1228547), FM by 1.18 meV atom$^{-1}$ under the recipe, is degenerate (0.001 meV atom$^{-1}$) once FM and AFM are compared in the same 90-atom cell; $TiFeBiPbO_6$ (mp-1216981), whose enumerated "AFM" cells had flipped only the $d^0$ Ti site, is AFM by 0.80–1.30 meV atom$^{-1}$ in all four Fe-only supercells; and $Tb_6FeBi_2$ (mp-1079285) and $FeP_6(WO_8)_3$ (mp-850225) remain FM, with same-cell margins on the enumerated AFM configurations (3.25 and 3.38 meV atom$^{-1}$) larger than the recipe numbers (2.32 and 2.09) suggested. In both compounds an ordering computed later for

Table 1. **Summary of DFT ordering verdicts for the 26 candidates.** Outcomes are classed by the fate of the net moment that the screening selects for. "Single Fe" counts candidates whose MP cell holds one Fe site, so that no Fe–Fe AFM order was representable in MP's calculation; "Tight" counts verdicts whose deciding margin comes from a same-cell FM/AFM pair (Methods). Each column counts candidates, not same-cell pairs; several candidates carry more than one pair.

| Outcome | Count | Single Fe | Tight |
|---|---|---|---|
| Net moment lost (AFM ground state) | 17 | 7 | 7 |
| Net moment kept (FM → FiM) | 3 | 1 | 1 |
| Consistent with label (2 FM, 2 FiM) | 4 | 2 | 4 |
| Degenerate ($\leq 0.10$ meV atom$^{-1}$) | 2 | 1 | 2 |
| Total | 26 | 11 | 14 |

the exchange model sets the final margin: the inter-chain AFM orderings of mp-1079285 lie 1.24–1.25 meV atom$^{-1}$ above FM, and a 68-atom cell of mp-850225 with alternate Fe layers antiparallel along the hexagonal $c$ axis lies 1.81 meV atom$^{-1}$ above FM. For mp-850225, "FM" describes the Fe sublattice, which is the only part the enumeration varies: its $W^{5+}$ moments (Methods) align antiparallel to the Fe, so the DFT ground state is strictly a collinear ferrimagnet, $2.0\,\mu_B$ per formula unit against $4.4\,\mu_B$ from Fe alone. Every non-FM verdict whose MP-recipe margin was below 2 meV atom$^{-1}$ was re-decided with a tight pair in the cell of its lowest valid non-FM configuration, and every one kept its sign, with margins moving by at most 0.32 meV atom$^{-1}$ (Supplementary Table 2); the one sign reversal came afterwards, when the metallic $Tb_6FeBi_2$ pair of mp-22972 was recomputed at a converged $k$-mesh (above); for $Li_4NbFe(WO_6)_2$ the tight pair ($-0.47$ meV atom$^{-1}$) lies just above a second, recipe-only configuration ($-0.54$ meV atom$^{-1}$), which therefore sets its margin. Fig. 2a marks which margins come from a same-cell pair.

The batch is drawn from the shortlist rather than from a random or stratified sample of MP's labelled population, so the loss rate above describes this structural family, not the database; given its oxide chemistry it is probably more AFM-prone than MP as a whole. What it does establish is that an MP net-moment label cannot be taken at face value for candidates drawn from exactly the population that DMI- and skyrmion-material screening

pipelines – including the one used here – select by construction.

### Most surviving net moments belong to hypothetical structures

A provenance check against MP's own metadata (`theoretical` flag and Inorganic Crystal Structure Database (ICSD) [32] cross-references, queried 2026-09-20) adds a caveat that applies to the shortlist as a whole rather than to the ordering verdicts. Twenty-four of the 26 candidates are flagged theoretical – structures generated by MP's substitution and prototype workflows, 23 of them with no ICSD entry at all (mp-551086 carries an ICSD identifier but is still flagged theoretical). Only mp-1197024 ($FeBi(SeO_3)_3$, DFT ground state AFM) and mp-22972 ($Tb_6FeBi_2$, five ICSD entries, FM Fe sublattice) are experimentally reported. Every other candidate that keeps a net moment is hypothetical, with MP hull distances of 0–85 meV atom$^{-1}$ (Fig. 2b), and one of them, mp-1079285, is a generated $Pm$ entry of the same compound as mp-22972: the two relax to the same structure (0.0085 Å rms) and give the same verdict (Methods). The funnel's symmetry and composition filters therefore preferentially select MP's generated structures, which is unsurprising – noncentrosymmetric cation orderings of otherwise centrosymmetric prototypes are exactly what substitution workflows produce – but it means a confirmed-FM verdict here is usually a statement about a candidate that may not be synthesizable. The one synthesized compound that keeps its moment, $Tb_6FeBi_2$, crystallizes in $P\bar{6}2m$ (point group $\bar{6}m2$), which has no Lifshitz invariant, so experimental follow-up for chiral textures would need a different candidate.

### First-principles exchange and DMI of two ferromagnetic candidates

For $Li_4Fe_2TeWO_{12}$ (mp-754090, space group $P1$), FM by 2.23 meV atom$^{-1}$ against the one AFM ordering its two-Fe primitive cell admits, and by 3.50 and 3.83 meV atom$^{-1}$ against layer orderings in cells doubled along $a$ and $c$ that also frustrate the same-sublattice bonds (Methods), the Fe sublattice is close-packed: each Fe has six neighbours in a triangular layer (5.11–5.16 Å) and six across the layer spacing (5.41–5.43 Å) (Fig. 3a). The same-cell FM/AFM pair gives one effective exchange for all twelve bonds, $J = 2.78$ meV per Fe–Fe bond ($zJ = 12J = 33.4$ meV), with an 8% protocol spread that we carry as its systematic uncertainty, and six pairs of constrained noncollinear canting probes resolve

the bond-averaged DMI vector, $\bar{\mathbf{D}} = (-0.018, 0.093, -0.123)$ meV per bond with $|\bar{\mathbf{D}}| = 0.155$ meV, so $|\bar{\mathbf{D}}|/J \approx 5.6\%$; the symmetric anisotropic exchange in the same probes is at most 0.025 meV (Methods). The single $J$ does not describe the doubled cells: it predicts $24J = 67$ meV per 40-atom cell for either layer ordering, against 140 and 153 meV from DFT. The same-sublattice bonds, which the primitive pair cannot see, are therefore ferromagnetic and stronger than $J$, and the single-$J$ model below underestimates the exchange stiffness of this compound. The DMI ratio is small and physically plausible: the magnetic ion at the probed bond, Fe, is a light $3d$ species with weak intrinsic SOC, and the tungsten that qualified this candidate for the funnel enters only through the ligand field. The presence of a heavy-SOC element anywhere in the composition does not guarantee a large DMI at the magnetic-ion bond that anchors a spin texture.

$Tb_6FeBi_2$ (mp-1079285, stored in space group $Pm$ but relaxing to the $P\bar{6}2m$ structure of mp-22972, one Fe per primitive cell) is a different magnetic topology: the Fe atoms form chains along $b$ (Fe–Fe 4.198 Å), with the chain next-nearest neighbours ($\pm 2b$) and six inter-chain neighbours all at 8.40–8.41 Å (Fig. 3b). Four same-cell pairs resolve three exchange shells. The chain pair places the chain-antiparallel state 58.43 meV per cell above FM, $J_b = 14.61$ meV. The two inter-chain pairs are nearly identical, $\Delta E_{2a} = 22.48$ and $\Delta E_{2c} = 22.24$ meV per 18-atom cell (1.25 and 1.24 meV atom$^{-1}$; Fe $\pm 1.64$–$1.67\,\mu_B$, induced Tb $\leq 0.21\,\mu_B$ in all four calculations), which supports the $J_a \approx J_c$ part of the mean-field assumption and gives $J_\perp = 2.79$ meV – a factor of 5.2 weaker than the chain coupling, but ferromagnetic, so the material is a three-dimensional ferromagnet of coupled FM chains rather than a quasi-one-dimensional one. The $1 \times 4 \times 1$ $++--$ pair places that state 84.92 meV per 36-atom cell above FM; subtracting $4J_b$ leaves $8J_{2b} = 26.50$ meV, i.e. $J_{2b} = 3.31$ meV – ferromagnetic and comparable to $J_\perp$, so the $\pm 2b$ bond is not a negligible correction: it carries 13% of $zJ = 2J_b + 2J_{2b} + 6J_\perp = 52.6$ meV and raises the mean-field estimate from $T_c^{\mathrm{MF}} = 178$ K (chain + inter-chain only) to 203 K. The chain-bond DMI vanishes by symmetry: in $P\bar{6}2m$ the Fe chain lies on the threefold axis and the midpoint of each chain bond on the $\sigma_h$ mirror, which forces $\mathbf{D} = 0$ (Moriya's rules). Spin spirals on our loosely relaxed $Pm$ geometry gave $|\mathbf{D}| = 0.098$ meV per chain bond ($|\mathbf{D}|/J_b = 0.67\%$), but that geometry departs from $P\bar{6}2m$ by up to 0.029 Å in the atomic positions and 0.1% in the cell, whereas the tight relaxation is $P\bar{6}2m$ to within 0.0065 Å; the nonzero $\mathbf{D}$ is a residue of the loose relaxation, not a property of the compound (Methods). At a converged $k$-mesh the

inter-chain margin rises from 1.25 to 1.31 meV atom$^{-1}$, which would raise $J_\perp$ by 5%, inside the spread quantified next, so the exchange model is kept.

$J_{2b}$ is a difference of two energies computed with different $k$-meshes, so we checked it with a same-cell pair: the $+-+-$ chain pattern in the $1 \times 4 \times 1$ cell, which flips all four chain bonds ($\Delta E = 8J_b$), gives $\Delta E = 100.45$ meV per 36-atom cell, i.e. $J_b = 12.56$ meV, 14% below the chain-pair value. The two pairs differ in cell shape and $k$-mesh (Methods), so the spread is the cell-shape and $k$-convergence sensitivity of a metallic Fe chain, and it brackets the exchange model: with the same-cell $J_b$ the residual is $J_{2b} = 4.34$ meV, $zJ = 50.6$ meV and $T_c^{\mathrm{MF}} = 196$ K, against $J_{2b} = 3.31$, $zJ = 52.6$ meV and 203 K with the chain-pair value (Fig. 3c, open markers). The production model uses the chain-pair parameters.

The third compound with a ferromagnetic Fe sublattice, $FeP_6(WO_8)_3$ (mp-850225), has two exchange shells, six in-plane 8.77 Å and six out-of-plane 8.91 Å Fe–Fe bonds. A second same-cell pair, with alternate hexagonal Fe layers antiparallel along $c$, flips every out-of-plane bond and no in-plane bond, so $\Delta E = 12J_2 = 123.2$ meV per 68-atom cell, giving $J_2 = 10.3$ meV and, with $8(J_1 + J_2) = 229.6$ meV from the first pair, $J_1 = 18.4$ meV, both ferromagnetic. The FM members of the two pairs (the same 68 atoms in different supercells) agree to 0.006 meV atom$^{-1}$. Because the $W^{5+}$ moments are free to follow the Fe, these are effective Fe–Fe couplings that include the W polarization. Its DMI could not be resolved: with one Fe per rhombohedral cell a two-sublattice canting probe sees no DMI at all, and of the spin-spiral pair that would resolve it in the 102-atom hexagonal cell, one chirality never reached self-consistency at the precision the extraction needs (Methods).

### Spin models order at 101 K and 139 K by finite-size scaling

The DFT exchange, with no fitting or rescaling, parameterizes classical Heisenberg models of the Fe sublattices that are run with Langevin spin dynamics in LAMMPS-SPIN [33] from the ordered state (Methods). Each model reproduces every DFT energy difference it was built from in a zero-step evaluation before any sweep.

For mp-754090, $|m|$ of the 2048-spin cell decreases monotonically from 0.966 at 10 K through 0.805 at 50 K and 0.659 at 75 K to 0.343 at 100 K, collapses to 0.068 at 125 K, and then sits on the finite-size floor (0.025–0.044 for 150–400 K; the $1/\sqrt{N}$ value for $N = 2048$ is 0.022) (Fig. 4a, Supplementary Table 3). An independent Metropolis Monte Carlo

calculation [34] of the same Hamiltonian on the same lattice gives $|m| = 0.967$, 0.912, 0.806, 0.661, 0.37±0.03 and 0.07±0.02 at 10, 25, 50, 75, 100 and 125 K, agreeing with the Langevin sweep to within 0.002 below 100 K and within the seed spread at 100 K, so the curve is the equilibrium $|m|(T)$ of the model and not an artifact of the thermostat. A control sweep that adds a DMI of the measured magnitude changes $|m|$ by at most 0.013 (at 100 K, within the seed spread) and by $\leq 0.005$ elsewhere, so an antisymmetric term of this size does not move the ordering scale of this model. The control used the $y$-projection measured first ($D = 0.097$ meV, 3.5% of $J$); the full bond-averaged vector, $|\bar{\mathbf{D}}| = 0.155$ meV, is 1.6 times larger and was not rerun.

Finite-size scaling locates the transition. With the box aspect ratio held fixed at $N = 256$, 864, 2048 and 4000 spins, the fourth-order Binder cumulant [35] $U_4 = 1 - \langle m^4\rangle/3\langle m^2\rangle^2$ of consecutive sizes crosses at 99.5, 101.3 and 102.7 K (Fig. 4b). The crossings drift upward monotonically with system size, the expected signature of corrections to scaling, giving $T_c = 101.2\pm1.3$ K, with the largest pair ($N = 2048\times4000$, least affected by those corrections) at 102.7 K; the susceptibility $\chi = N(\langle m^2\rangle - \langle|m|\rangle^2)/k_BT$ peaks at 105 K independently at all four sizes. The transition of this model is therefore the classical-Heisenberg fcc value, $k_BT_c \approx 3.18\,J$ or $\approx 103$ K [37], to within the 5 K grid spacing, while mean field ($T_c^{\mathrm{MF}} = zJ/3k_B = 129$ K) overestimates it by 28%. The 8% protocol spread in $J$ shifts the transition by about 8 K.

For the three-shell mp-1079285 model, $|m|$ falls smoothly from 0.988 at 5 K to 0.672 at 100 K and 0.509 at 125 K, then collapses to 0.128 at 150 K (the one point with a seed-to-seed spread, ±0.012, as expected at the transition of a finite cell) and sits on the finite-size floor of 0.03–0.05 from 200 K (Fig. 4c, Supplementary Table 4). The same-cell parametrization, swept at 10, 100, 125, 150 and 200 K, gives $|m| = 0.975$, 0.668, 0.497, 0.124 and 0.047 against 0.975, 0.672, 0.509, 0.128 and 0.048 for the production model, so the 4% change in $zJ$ does not move the collapse from between 125 and 150 K. The sweep was run with the loose-geometry DMI ($|\mathbf{D}|/J_b = 0.67\%$), which vanishes in the symmetric structure; its canting angle, of order $D/2J \approx 0.2°$ per bond, is far below the thermal disorder at any temperature on the grid, so removing it does not change the sweep.

The $U_4(T)$ curves of consecutive sizes cross at 140.9, 141.4 and 135.6 K, giving $T_c = 139.3 \pm 2.6$ K, and $\chi$ peaks at 145 K for the three smaller cells and 140 K for the largest (Fig. 4d). This puts the transition at $0.68\,T_c^{\mathrm{MF}}$. Each Fe here has $2+2+6 = 10$ neighbours

on three shells, so no textbook lattice applies exactly; the ratio lies just below the isotropic simple-cubic ($z = 6$) value 0.72 [36] and the fcc ($z = 12$) value 0.79 [37], as the strong exchange anisotropy leads one to expect. The crossings are not monotonic in system size as they are for mp-754090, and the analysis is harder here: the exchange is strongly anisotropic ($J_b/J_\perp = 5.2$, though ferromagnetic on every shell), so intra-chain correlations survive well above the three-dimensional ordering temperature and the $U_4$ curves separate gradually rather than pivoting about a point, and the fluctuations near $T_c$ are correspondingly larger. The 135–150 K window was therefore run with 10–14 seeds per point, six of them 80 ps; the three crossings then agree to ±2.6 K. Both spreads over pairwise crossings are lower bounds on the uncertainty of the model $T_c$: re-deriving the crossings under changes to the sampling window shows that the autocorrelation time of $U_4$ near $T_c$ is comparable to the run length (Methods), and the DFT coupling uncertainties come on top.

## DISCUSSION

Fahmy's classifier [16] detects the Materials Project's ferromagnetic bias statistically, at large $N$; the present work reaches the same conclusion by the opposite route, a small-$N$ DFT energy-mapping check, in exactly the noncentrosymmetric, heavy-SOC-element, net-moment slice that DMI- and skyrmion-material pipelines target. That 17 of 26 candidates lose their net moment in that slice (13 of 21 distinct compositions) is a direct warning to any workflow, including our own, that uses MP's ordering label as a hard screening criterion rather than a prior. The finding is not new in kind – MP documents that FM-initialized entries can miss AFM ground states [15] – and much of it is explained by the Fe-oxide chemistry the filters select, but it is large in exactly the population where the label is used as a filter.

The consequence runs in both directions. The funnel used here excludes AFM-labeled candidates, on the reasoning that a net moment is required for the texture-hosting mechanisms it targets. If most shortlisted net-moment candidates are in fact compensated antiferromagnets, downstream compute – structure backfill, potential relaxation, spin dynamics – risks being spent on candidates whose net-moment premise is wrong, while part of the true net-moment landscape may be filtered out before any calculation is spent on it. The corrective is not to trust a classifier over MP either – both are estimates – but to treat any ordering

field, database-recorded or predicted, as a prior to be checked with DFT before committing expensive downstream stages. The provenance result adds a second, independent prior: in this family the screening preferentially returns hypothetical structures (24 of 26 here, and six of the seven candidates that keep a net moment), so a verified magnetic ground state still has to be paired with a synthesizability assessment before it is an experimental target.

For two compounds whose Fe sublattice remains FM, the full chain from DFT to finite temperature costs one and five same-cell pairs (plus two doubled-cell orderings for the first) and twelve and four constrained noncollinear calculations, respectively (the four on $Tb_6FeBi_2$ now serving only to show that its DMI is a geometry residue), plus a finite-size-scaling campaign, and returns ordering temperatures of the classical models near 100 K and 140 K with no adjustable parameter. The quoted Binder-crossing spreads ($\pm 1$ and $\pm 3$ K) are statistical only; the 8% and 14% spreads of the DFT exchange constants (Results) move them by about 8 K and 20 K, and the classical treatment of the moment is a larger, systematic uncertainty. The one nonzero DMI-to-exchange ratio is a few percent, and the other vanishes by symmetry, despite the heavy elements that qualified the compounds for screening, which argues for adding a magnetic-ion-resolved SOC criterion to DMI screening rather than an element-presence one.

Several limitations bound these conclusions. The 26-candidate batch consists of the Materials Project entries in one DMI-filtered shortlist, not a random sample of MP's labelled population (1,864 candidates pass the DMI funnel, 1,195 with the hull as a hard filter), so the loss rate is a property of this structural family. Only collinear orderings were enumerated, so a spiral or canted ground state – the state a strong-DMI compound would adopt – cannot be identified, and "consistent with the label" means lowest among the collinear states tested. The verdict rule is deliberately asymmetric, and the 12 non-FM verdicts decided on an MP-recipe pair alone carry a protocol uncertainty that the sixteen doubly-computed configurations bound at 1.3 meV atom$^{-1}$; only one of them has $|\text{margin}| < 2$ meV atom$^{-1}$ ($Li_4NbFe(WO_6)_2$, $-0.54$), and a tight pair on a second configuration corroborates its sign ($-0.47$ meV atom$^{-1}$). The mp-754090 probes resolve only the bond average of what is, in $P1$ symmetry, eight unconstrained three-component DMI vectors, so its spin model is exchange-only; the DMI of mp-850225 is not resolved. The spin models are classical, unit-spin, rigid-lattice models. For $S = 5/2$ the quantum and classical mean-field ordering temperatures differ by the factor $S(S+1)/S^2 = 1.4$, and the energy mapping here absorbs

$S^2$ into $J$, so depending on how the moment is quantized the classical $T_c$ can be low by up to that factor; the itinerant Fe moment of $Tb_6FeBi_2$, which changes by 10% between its FM, AFM and spiral states, is a further departure from a rigid-moment model. The spin models also have zero single-ion anisotropy, which was not probed in any compound; the mp-754090 model assigns one $J$ to two geometrically distinct shells, the mp-1079285 model one $J_\perp$ to six inter-chain bonds, and Tb is treated with its $4f$ electrons in the core, so any Tb–Fe exchange is absent. The Binder-cumulant crossings are therefore critical temperatures of the spin models, not of the materials.

Within the structural family that DMI-candidate screening selects, MP's net-moment label survives a collinear DFT test for fewer than a third of the candidates. High-throughput ordering labels should be used as a screening prior rather than ground truth, particularly in the structural families that a downstream physical mechanism – here, the DMI – selects for.

## METHODS

### Harmonized multi-source database

Materials Project [14] (37,783 records, via the `mp-api` REST interface [38]), JARVIS-DFT [19] (19,116), AFLOW [20] (2,639), MAGNDATA [17, 18] (1,919 of 2,414 attempted), 2DMatPedia [21] (1,862), OQMD [22, 23] (816), and NOMAD [24] (500) were merged into one schema (`material_id`, `formula`, `elements`, `space_group`, `density`, `volume`, `band_gap`, `ordering`, `total_magnetization`, and, where available, full structure and per-site magnetic-moment vectors), for 64,635 total records. Records retain a source tag rather than being silently deduplicated across sources. MAGNDATA is the only source with experimentally determined (rather than DFT-derived) magnetic structure, and is accordingly the ground-truth source for classifier training and validation. A scraper against the live Bilbao Crystallographic Server index recovered 1,919 of 2,414 entries (79.5%); the failures, mostly transient server errors, are analysed in Supplementary Note 1. Ordering was validated on a known case ($LaMnO_3$, correctly recovered as AFM, space group $Pnma$).

### Structure-aware magnetic-ordering classifier

A LightGBM [25] gradient-boosted-decision-tree classifier predicts ordering (FM, AFM, FiM or nonmagnetic, NM) from composition- and structure-derived features, in the spirit of general-purpose composition-based property models [39, 40] and of earlier machine-learning treatments of magnetic order and ordering temperature [41–44]. A composition-only baseline (per-element statistics of atomic number, mass, electronegativity, atomic radius, plus group, period, transition-metal and rare-earth fractions) achieves 56.7% accuracy (60.7% balanced accuracy). Adding two tiers of structural features – broad, near-universal-coverage fields (space group number, crystal-system bucket, unit-cell volume, density, site count) and local-geometry features around magnetic sites (nearest magnetic–magnetic neighbour distance, magnetic coordination number, fraction of magnetic sites; computed from atomic positions and species only, never from the magnetic-moment vectors that generate the label itself) – raises this to 67.4% accuracy (68.0% balanced accuracy) on a stratified 28,769/7,193 train/test split (Supplementary Fig. 1). Because records are not deduplicated across sources, the same compound can sit on both sides of the split. Of the 7,193 test records (6,791 MP, 401 MAGNDATA, 1 AFLOW), 37.8% share a reduced formula with a training record, but only 2.5% share formula and space group, and 1.6% formula, space group and label; the structural near-duplicates are concentrated in MAGNDATA (34.9% of its test records, against 0.6% for MP), so the MAGNDATA test accuracy is the figure most inflated by leakage. The local-geometry tier is computed for every source with a public structure endpoint (Materials Project 37,783/37,783, JARVIS 19,116/19,116, AFLOW 2,637/2,639, NOMAD 495/500, 2DMatPedia 1,862/1,862; OQMD 0/816 because its REST endpoint was unreachable). Computing it for MAGNDATA alone, whose CIFs carry atomic positions inline, would make "tier present" a proxy for the one source whose labels are experimental and 83% AFM; with full coverage the headline numbers are essentially unchanged (67.0% $\rightarrow$ 67.4% accuracy, 68.1% $\rightarrow$ 68.0% balanced), but per-source test accuracy on Materials Project rises from 65.6% to 66.9% (balanced 63.0% $\rightarrow$ 65.9%), the inflated MAGNDATA figure drops from 89.5% to 75.8%, and four local-geometry features (mean/std/min magnetic–magnetic distance, mean magnetic coordination) enter the top seven features by LightGBM split importance, where before none appeared in the top ten. Space-group number remains the single most important feature, despite being absent from the baseline entirely. The classifier

is retrained on all labeled data and deployed to predict ordering for every record lacking a ground-truth label, including all 19,116 JARVIS records, none of which carry a populated ordering field.

### Graph neural network classifiers

Two graph-based models were trained on the train/test split of the LightGBM classifier and scored, together with it, on the 7,161 of its 7,193 test records whose structure is fully ordered (the other 32 are partially occupied MAGNDATA structures). Each crystal is a periodic graph whose nodes are atoms and whose edges join every atom to its 12 nearest neighbours within 8 Å, built from species, positions and lattice only; no magnetic moment enters, since the MAGNDATA labels are derived from the moments. The first is a crystal graph convolutional neural network (CGCNN) [45] trained from scratch in PyTorch Geometric [46]: a learned 64-dimensional element embedding projected to 128 channels, four gated graph convolutions with batch normalization and residual updates, bond lengths expanded in 40 Gaussians over 0–8 Å, mean pooling over atoms and a two-layer head. It is trained with class-weighted cross-entropy and AdamW (learning rate $10^{-3}$, weight decay $10^{-4}$, cosine schedule) for 120 epochs in batches of 128, keeping the epoch of best balanced accuracy on a stratified 10% validation fraction of the training set, with three seeds on one 24 GB GPU. The second probes a pretrained foundation potential: the invariant per-atom descriptors of both interaction layers of the frozen MACE-MP-0 `medium` model [47] (128 channels, `max_L`= 1, $r_{\max} = 6.0$ Å; distinct from the fine-tuned `small` model used for relaxations below) are pooled over atoms by mean and standard deviation into a 512-dimensional vector, which LightGBM, with the settings of the screening classifier, classifies alone and concatenated with the screening features. MACE-MP-0 is trained on energies, forces and stresses only, so the embedding carries no ordering label, but it is not independent of Materials Project: its MPtrj training set holds relaxation trajectories of the same structures, computed in their stored magnetic orderings. Accuracy on the Materials Project rows measures agreement with Materials Project's label, which is the label this work finds biased towards FM; the MAGNDATA rows (311 AFM, 39 FiM, 19 FM in the test set) are the only experimental ground truth, and their balanced accuracy shifts by up to 4 points when the training set changes by only the 26 candidates and by 6 points between CGCNN seeds,

as much as it differs between models. For an external test against DFT, every model was retrained with the 26 DFT-validated candidates removed from training, and its predictions for them compared with the DFT verdicts; with 21 non-FM and 3 FM verdicts these are reported as counts.

**Screening funnel and shortlist**

The noncentrosymmetric filter was verified computationally against all 230 space groups, recovering the textbook 92 centrosymmetric / 138 noncentrosymmetric split. A hull energy is known for 1,340 of the 1,864 heavy-SOC survivors (the MP, JARVIS and OQMD records), 671 of which lie within 100 meV atom$^{-1}$ of it. The `ehull` field of the JARVIS-DFT snapshot used here is not a hull distance (Supplementary Note 2), so the hull distance of every JARVIS record was recomputed from JARVIS's own OptB88vdW formation energies with a per-chemical-system convex hull (13,967 of 19,116 records within 100 meV atom$^{-1}$); three of the four JARVIS entries in the original top-30 shortlist fail this criterion, and none entered the DFT batch, which is Materials Project only. The hull criterion is available both as a hard filter (records from sources that provide no hull energy – AFLOW, NOMAD, 2DMatPedia, MAGNDATA – are retained rather than penalised for a missing field; 1,195 candidates) and as a sort key; the DFT batch was drawn from the sort-key ranking, which is kept alongside so the batch remains reproducible.

For the top 100 screened candidates per source, structures were backfilled via each source's own public data-access method (298 of 300 attempted backfills succeeded; two AFLOW records failed on repeated TLS errors from the AFLUX endpoint) and re-relaxed with a MACE foundation-model potential [47, 48], fine-tuned by multihead training on 42,610 MPtrj frames (38,349 training, 4,261 validation) of the Materials Project materials in the harmonized set, with a replay head sampling 20,000 configurations from the original pre-training distribution to avoid catastrophic forgetting. The harmonized database itself holds no energies or forces, and both the base model and the fine-tune are trained on Materials Project calculations, so the relaxation is a consistency check against each source's own structure, not an independent level of theory. The base model is MACE-MP-0 `small` (`num_channels`= 128, `max_L`= 0, two interaction layers, $r_{\max} = 6.0$ Å); the relaxations use the Stage-2 stochastic-weight-averaging checkpoint of the fine-tune with its default head, and

are static FIRE [49] optimizations in ASE [50]. Other universal potentials trained on Materials Project data, such as M3GNet [51] and CHGNet [52], would serve the same purpose. All 298 relaxations converged (max force $\leq 0.05$ eV Å$^{-1}$) with a median relaxation-energy shift of 0.9 meV atom$^{-1}$ relative to the source structure. Classifier confidence and MACE relaxation consistency were merged into a final ranking (converged $\rightarrow$ hull-stable $\rightarrow$ smallest relaxation shift $\rightarrow$ classifier confidence). The DFT batch is the 26 Materials Project entries of the resulting top-30 shortlist; the other four are JARVIS records, three of which fail the recomputed hull criterion. Because this ranking puts convergence and relaxation consistency ahead of hull distance, the 26 sit anywhere from rank 1 to rank 663 of the hull-sorted ranking (Supplementary Fig. 2).

### DFT ordering-validation protocol

For shortlisted Materials Project candidates, whose ordering field holds MP's own computed ordering rather than a classifier prediction (our harmonized schema marks such labels `ordering_source=ground_truth`, `confidence=1.0`; these are fields of our schema, not of MP's), we apply the collinear magnetic-ordering energy-mapping approach behind MP's own workflow [15]: symmetry-distinct FM and AFM spin configurations are enumerated with `pymatgen.MagneticStructureEnumerator` [53] (backed by `enumlib`); VASP inputs are generated with `MPRelaxSet` – MP's own PBE exchange-correlation functional [54], GGA+U parameters [55–57] (e.g. LDAUU= 5.3 eV on Fe), plane-wave cutoff (520 eV), and Monkhorst–Pack $k$-point convention [58] – overriding only the initial MAGMOM per configuration; and each configuration is relaxed to convergence within Kohn–Sham density-functional theory [59, 60] as implemented in VASP 6.4.3 [61] with the projector-augmented-wave method [62, 63] (`MPRelaxSet` defaults ISIF= 3, ISMEAR= $-5$, i.e. the tetrahedron method with Blöchl corrections [64], EDIFF= $5 \times 10^{-5}$ eV per atom, i.e. 0.05 meV atom$^{-1}$). This keeps the energies directly comparable to MP's stored values rather than constituting an independent DFT recipe. Beyond VASP's own convergence flag, every calculation was audited in two further ways: the final RMS charge-density residual was checked (the "EDIFF is reached" banner can fire on a transient while the density is still far from self-consistent), and the final per-ion moments were compared against the intended MAGMOM pattern and against the moment magnitudes of the FM calculation of the same compound. A configu-

ration whose magnetic ions collapse to a low-spin or nonmagnetic state, or whose moments re-order into the FM pattern during the self-consistent-field (SCF) cycle (a doubled "AFM" cell that is really FM), passes VASP's convergence test but is not a sample of the ordering it was labeled with, and is excluded from the verdict rather than counted. As a check of the setup, the converged FM configuration of mp-861559 ($TaFeO_4$) reproduces MP's recorded total magnetization to within $0.003\,\mu_B$ (30.00000 vs. $30.0022\,\mu_B$ for 36 atoms), ruling out a convention mismatch, while three symmetry-distinct AFM configurations all converge below it, the lowest by 0.274 eV per cell (≈45.7 meV per Fe).

The first pass (93 relaxations across the 26 candidates; 116 audited relaxations in total once the second batch is included) left 12 calculations in seven candidates that failed the spin-state audit: collapsed or low-spin Fe/Cr moments in an AFM configuration (mp-1173139, mp-1173143, mp-551086, mp-775194), "AFM" cells whose moments re-ordered to the FM pattern during the SCF cycle (mp-682554), an "AFM" pattern that had flipped only a nonmagnetic $d^0$ Ti site and was therefore FM in disguise (mp-1216981), and an FM reference that collapsed to a nonmagnetic state (mp-22972). A second batch replaced every failed configuration with a freshly enumerated one in a spin state that the audit accepts: additional symmetry-distinct AFM/FiM cells for mp-1173139, mp-1173143, mp-775194 and mp-551086 (the ferrimagnetic ones with the two Fe sublattices, or Fe and Cr, unequal in number), Fe-only AFM supercells for mp-1216981 (Ti excluded from the enumeration), anti-collapse FM reruns for mp-22972 (in the primitive cell and in the cell of its lowest AFM configuration), and, for mp-682554, whose $1 \times 1 \times 2$ Fe–Fe stacking (130 atoms) diverged twice at the first density-mixing step through charge sloshing, five single-point mixer/smearing variants of the same cell run side by side, with the production relaxation restarted only under the first variant that converged; that production run was stopped before completion and is the one configuration missing from the batch. A superseded attempt (a diverged or collapsed run that was re-done) is neither a sample nor a gap in the count.

Two protocol effects matter at the sub-meV atom$^{-1}$ level. First, the MP recipe compares an FM relaxation in the primitive cell with AFM relaxations in supercells, each with its own $k$-mesh and each relaxed with ISIF= 3 to its own cell shape; for the sixteen configurations run both ways this "loose" margin differs from a "tight" one (both orderings relaxed with ISIF= 2 in the same relaxed AFM cell, identical $k$-mesh, EDIFF= $10^{-6}$ eV, EDIFFG= $-0.01$ eV Å$^{-1}$) by up to 1.3 meV atom$^{-1}$ in either direction ($Ba_4Ta_{10}FeO_{30}$: $+1.18 \rightarrow 0.00$; $FeP_6(WO_8)_3$:

+2.09 → +3.38). Every candidate whose verdict rested on a loose margin smaller than about 2 meV atom$^{-1}$ was therefore re-decided with a tight pair built in the cell of its lowest valid non-FM configuration (14 pairs in the first tight batch, 6 more for the remaining small-margin non-FM verdicts), and a tight number replaces the loose one for that configuration wherever it exists. Second, the verdict rules are asymmetric by construction: a non-FM verdict is reached as soon as one audited configuration lies more than 0.10 meV atom$^{-1}$ below FM – a still-missing configuration cannot undo that – whereas an FM verdict and *degenerate* require every staged configuration to have a valid energy. The 0.10 meV atom$^{-1}$ window is twice the per-atom EDIFF of the loose protocol. The two 130-atom tight relaxations of mp-682554, first cut off short of the force criterion, were continued to convergence (29 and 32 ionic steps, final maximum forces 0.006 and 0.010 eV Å$^{-1}$); the gap moved from 0.315 to 0.313 meV atom$^{-1}$ (40.7 meV per cell). Three further biases are small but one-sided or uncontrolled. Tight pairs are built in the relaxed cell of the AFM configuration, which favours AFM slightly, so a tight FM verdict is conservative and a tight AFM verdict slightly generous. `MPRelaxSet` keeps the tetrahedron method (ISMEAR= −5) during relaxation, whose forces are inaccurate for metals; MP stores a zero band gap for mp-754090, mp-551086, mp-1218521 and mp-1079285, and mp-682554 and mp-22972 are metallic too, but both orderings of a pair use the same smearing and $k$-mesh, so the error enters the margin only through the small structural difference between them. Finally, only $U_{\mathrm{Fe}} = 5.3$ eV was used; since antiferromagnetic superexchange scales as $t^2/U$, the size, though not the sign, of the oxide margins depends on this choice. The numerical precision of a same-cell margin was checked on the two smallest tight pairs, $CaLaTaFeO_6$ (mp-1043970) and $Zn_2FeWO_6$ (mp-1047414). Static recalculations of both relaxed configurations of each pair used a $k$-mesh 1.5 times denser in each direction and a 650 eV plane-wave cutoff (520 eV in production). The margins moved from −0.0975 to −0.0975 meV atom$^{-1}$ and from −0.1852 to −0.1842 meV atom$^{-1}$ (AFM−FM), a shift of $\leq 0.001$ meV atom$^{-1}$. $CaLaTaFeO_6$ is therefore numerically AFM at this level of theory. It stays in the degenerate class because its margin lies inside the 0.10 meV atom$^{-1}$ window, which we keep for the $U$ and functional sensitivity discussed above rather than for numerical noise. Both of these pairs are insulating. In a metal the $k$-mesh is not a small perturbation: at MP's $k$-point density the tight pair of the metallic $Tb_6FeBi_2$ entry mp-22972 had the wrong sign, which a uniform ∼60 Å $k$-length corrected (see "Exchange, symmetry and $k$-convergence of $Tb_6FeBi_2$"), whereas the metallic

mp-682554 margin moved by only 0.004 meV atom$^{-1}$ on a mesh 3-fold denser along each axis and kept its sign.

Eight candidates contain tungsten. Three of them ($Zn_2FeWO_6$, $Ba_3Fe_2WO_9$ and $Ca_3Fe_2(WO_6)_2$) converged under MP's standard Hubbard correction on W (LDAUU= 6.2 eV); the other five (mp-754090, mp-755203, mp-769471, mp-775194 and mp-850225) did not converge with our settings, and were run with no $U$ on tungsten (LDAUL= $-1$), applied uniformly across every ordering of each compound, which converged every corresponding production calculation. These reruns also fix the cell shape and volume at MP's relaxed values (ISIF= 2, ions only) and use Gaussian smearing (ISMEAR= 0, $\sigma = 0.05$ eV; tetrahedron smearing for mp-755203) for SCF stability. This changes the absolute energy scale of those five candidates relative to MP's U-inclusive value and is a departure from MP's recipe for mp-754090 and mp-850225, two of the three candidates whose Fe sublattice stays FM. For a nominally $d^0$ $W^{6+}$ the FM-versus-AFM comparison is insensitive to $U_W$, since the same settings apply to every ordering. $FeP_6(WO_8)_3$ (mp-850225) is the exception: charge balance with $Fe^{3+}$ and $P^{5+}$ requires $W^{5+}$ ($d^1$), and in the U-free runs each W carries 0.78–0.86 $\mu_B$ antiparallel to Fe in the FM state, giving 2.0 $\mu_B$ per formula unit against the 8 $\mu_B$ MP stores for W parallel to Fe. A rerun at MP's $U_W = 6.2$ eV on MP's 34-atom structure, started once with W parallel and once with W antiparallel to Fe and using linear charge mixing (AMIX= 0.1), failed in the same way as the earlier attempts: in both, the residual charge density grew above 100 within 30 self-consistent steps after the fixed-density start. Whether MP's 8 $\mu_B$ state is the ground state of MP's own functional therefore remains untested; the ferrimagnetic assignment of mp-850225 holds for $U_W = 0$ only.

**Exchange and DMI of mp-754090**

$Li_4Fe_2TeWO_{12}$ is a hypothetical $P1$ cation ordering of a rock-salt-derived oxide. With $Li^+$, $Te^{6+}$, $W^{6+}$ and $O^{2-}$ charge balance requires $Fe^{4+}$; in the tight FM calculation each Fe carries 4.16 $\mu_B$, between the $Fe^{3+}$ and $Fe^{4+}$ values, the oxygen sublattice carries $-0.45$ $\mu_B$ in total, and the gap is below 0.01 eV (MP also stores a zero gap). The hole is therefore shared between Fe and O, as in the negative-charge-transfer ferrates, and the compound is a metal or near-metal in this description. Its thermodynamic and cation-order stability have not been tested, and a real Li/Fe/Te/W rock-salt is likely to be partly cation-disordered.

Effective spin-Hamiltonian parameters are extracted by energy mapping, the standard route from the total energies of constructed spin configurations to Heisenberg parameters [65–67]. The isotropic exchange is obtained from the tight same-cell FM/AFM pair used for the ordering verdict (both orderings relaxed in the same cell, which is the fixed lattice a Heisenberg $J$ is defined on). Each of the two Fe sites (Fe0, Fe1) per primitive cell of the rock-salt-derived $Li_4Fe_2TeWO_{12}$ structure has twelve Fe neighbours, six in a triangular layer (5.112–5.157 Å, bond angles of 60°, 120° and 180°; four on the opposite sublattice and two on the same sublattice by lattice translation) and six across the 4.53 Å layer spacing (5.411–5.427 Å; again four on the opposite sublattice and two on the same). The next shell starts at 7.45 Å. For $H = -\sum_{\langle ij \rangle} J\, \hat{s}_i \cdot \hat{s}_j$ with each bond counted once, the collinear AFM configuration (Fe0 sublattice reversed) flips the eight inter-sublattice bonds per cell – four in-plane and four inter-layer – so $E_{\mathrm{AFM}} - E_{\mathrm{FM}} = 2 \times 8\, J = 16J$; with the same-cell DFT value $\Delta E = 44.54$ meV per two-Fe cell this gives $J = \Delta E/16 = 2.78$ meV per bond. The loose pair (each ordering at its own ISIF= 3 optimum) gives $\Delta E = 48.23$ meV and $J = 3.01$ meV, an 8% protocol spread that we carry as the systematic uncertainty of $J$. A single FM/AFM pair cannot separate the in-plane from the inter-layer exchange, nor resolve the four same-sublattice bonds, which are invisible to a two-configuration fit; all twelve bonds are assigned the same $J$ in the spin model – an explicit assumption. This is a single-$J$ effective model, not a multi-shell decomposition; its $zJ = 12J = 33.4$ meV, which fixes the mean-field ordering temperature, is what the DFT pair determines directly.

To test the FM verdict against orderings the primitive cell cannot hold, we doubled the relaxed tight FM cell along $a$ and along $c$. In each 40-atom cell we compared FM with a layer ordering, the two primitive-cell images antiparallel. These are static calculations at the tight settings ($U_{\mathrm{W}} = 0$, $k$-mesh halved along the doubled axis). The layer orderings lie 140.1 ($2a$) and 153.0 meV ($2c$) per cell above FM, i.e. 3.50 and 3.83 meV atom$^{-1}$. Both FM twins agree with twice the primitive FM energy to within 2 meV per cell. The layer orderings flip all four same-sublattice bonds along the doubled axis and eight of the sixteen inter-sublattice bonds. If only the twelve nearest neighbours couple and the inter-sublattice $J$ is held at 2.78 meV, the same-sublattice bonds along $a$ and $c$ come out at about 12 and 14 meV, ferromagnetic. The AFM cells were not relaxed, which overstates both costs by their relaxation energy, and the split depends on the twelve-neighbour assumption. The robust results are that FM stays lowest against both orderings, and that the single-$J$ model underestimates the ferromagnetic

coupling; its ordering temperature is therefore a low estimate for this structure.

DMI does not appear in a collinear FM-versus-AFM comparison; it requires spins canted relative to one another. The Li/Fe/Te/W cation ordering reduces the space group to $P1$, so no inversion centre at a bond midpoint forbids it [2]. We ran two static, constrained noncollinear-spin+SOC calculations from the relaxed FM structure – the established route to a first-principles DMI [68, 69] – within unconstrained noncollinear PAW magnetism [70], using the penalty-energy method [71] (`I_CONSTRAINED_M=1`, constraining moment direction only) to hold Fe0 along $+\hat{z}$ and Fe1 along $+\hat{x}$ (canted$_+$) or $-\hat{x}$ (canted$_-$) – i.e. $\theta = \pm 90°$ canting in the $xz$-plane, an arbitrary but fixed reference axis since $P1$ symmetry privileges no plane. The constraint held essentially exactly (Fe1 converged to $4.038\,\mu_B$ along $x$ in canted$_+$ and $-4.039\,\mu_B$ in canted$_-$; Fe0 leakage off $z$ was $< 0.002\,\mu_B$ in both). With the same bond-counting convention as for $J$ ($z = 8$ inter-sublattice bonds per cell, each counted once; the four same-sublattice bonds connect parallel spins and contribute to neither term), $E(\theta) = E_0 - zJ\cos\theta - zD\sin\theta$, and comparing $\theta = \pm 90°$ isolates the antisymmetric term since $\cos(\pm 90°) = 0$:

$$zD = \frac{E(\theta{=}-90°) - E(\theta{=}+90°)}{2}, \tag{1}$$

giving $E(-90°) - E(+90°) = 1.546$ meV per cell and $D = 0.097$ meV per bond, i.e. $D/J \approx 3.5\%$, if the symmetric part of the exchange tensor is neglected. Because $\hat{s}_1 = \hat{z}$ and the canting is confined to the $xz$-plane, $\hat{s}_1 \times \hat{s}_2 = (0, \sin\theta, 0)$: this probe measures the mean $y$-component of the DMI vectors of the eight inequivalent inter-sublattice bonds in this lab frame, not a generic magnitude, and is a resolved DFT number for one projection of $D$ rather than a fully characterized three-dimensional DMI vector.

A single canted pair cannot separate the antisymmetric term from the symmetric anisotropic exchange $\Gamma$ (the traceless symmetric part of the bond-summed $3 \times 3$ exchange tensor $\mathcal{T}$, with $E = -\hat{s}_0 \cdot \mathcal{T}\hat{s}_1$ summed over the eight bonds). Swapping the roles of the two Fe does: for orthogonal axes $\hat{a}, \hat{b}$, let $\Delta_{ab} = E(\hat{s}_0{=}\hat{a}, \hat{s}_1{=}{+}\hat{b}) - E(\hat{s}_0{=}\hat{a}, \hat{s}_1{=}{-}\hat{b})$. Then

$$\bar{\mathbf{D}} \cdot (\hat{a} \times \hat{b}) = -\frac{\Delta_{ab} - \Delta_{ba}}{4z}, \qquad \bar{\Gamma}_{ab} = -\frac{\Delta_{ab} + \Delta_{ba}}{4z}, \tag{2}$$

per bond, with $z = 8$. We ran the reversed $xz$ pair and both orderings of the $yz$ and $xy$ pairs (ten further constrained calculations, same settings, all converged to EDIFF $= 10^{-7}$ eV). The energy differences are $\Delta_{zx} = -1.546$, $\Delta_{xz} = +1.420$, $\Delta_{zy} = +0.112$, $\Delta_{yz} = +0.698$, $\Delta_{xy} = +2.065$ and $\Delta_{yx} = -1.882$ meV per cell. They give $\bar{\mathbf{D}} = (-0.018, 0.093, -0.123)$ meV,

$|\bar{\mathbf{D}}| = 0.155$ meV and $|\bar{\mathbf{D}}|/J = 5.6\%$, and off-diagonal $\bar{\Gamma}_{zx} = 0.004$, $\bar{\Gamma}_{yz} = -0.025$ and $\bar{\Gamma}_{xy} = -0.006$ meV. The first probe was 96% $D_y$, but it missed the largest component, $D_z$. $\bar{\mathbf{D}}$ is the average of the eight bond vectors. It sets the relative canting of the two Fe sublattices. It is not the bond-resolved set that enters a spiral or skyrmion energy, which this geometry cannot isolate (next paragraph).

### Exchange, symmetry and $k$-convergence of $Tb_6FeBi_2$

The Fe atoms of $Tb_6FeBi_2$ form chains along $b$ (Fe–Fe 4.198 Å), the next Fe–Fe distances being 8.396 Å along $\pm 2b$ and 8.396–8.407 Å to the six chain neighbours along $\pm a$, $\pm c$, $\pm(a+c)$. Tb is treated with the $4f$ electrons frozen in the core (`Tb_3` PAW), so the only moment in the model is Fe ($1.68\,\mu_B$ in the FM calculation), an explicit approximation that ignores any Tb–Fe exchange. Every collinear AFM configuration that the enumeration produced for this candidate turned out to be the same chain-antiparallel order in a different cell setting, so the tight pair in the $1 \times 2 \times 1$ cell resolves only the intra-chain coupling: with $H = -\sum_{\langle ij \rangle} J_{ij}\, \hat{s}_i \cdot \hat{s}_j$ (each bond counted once) it flips two chain bonds per cell, $E_{\mathrm{AFM}} - E_{\mathrm{FM}} = 4J_b$. A chain model alone has no order at any $T > 0$, so three further same-cell tight pairs were staged in supercells of the relaxed FM cell (`ISIF`= 2, `EDIFF`= $10^{-6}$ eV, `EDIFFG`= $-0.01$ eV Å$^{-1}$): the $2a$ and $2c$ cells with the two Fe antiparallel, which flip the $\pm a, \pm(a+c)$ and the $\pm c, \pm(a+c)$ bonds respectively ($\Delta E_{2a} = 4(J_a + J_{ac})$, $\Delta E_{2c} = 4(J_c + J_{ac})$), and a $1\times4\times1$ cell with the chain pattern $++--$, which flips two of the four chain bonds and all four $\pm 2b$ bonds ($\Delta E_{4b} = 4J_b + 8J_{2b}$). Two equations do not determine three inter-chain couplings; the model takes their mean, $J_\perp = (\Delta E_{2a} + \Delta E_{2c})/16$, under the stated assumption $J_a \approx J_c \approx J_{ac}$ (the three bond lengths agree to 0.01 Å). The $\pm 2b$ bond is resolved separately because its length coincides with the inter-chain shell, which no radial exchange function in the spin-dynamics code can separate. The $J_b$ and $\Delta E_{4b}$ pairs use different $k$-meshes ($3\times3\times3$ on the $1 \times 2 \times 1$ cell, $3 \times 2 \times 3$ on the $1 \times 4 \times 1$ cell) and cells: the $1 \times 2 \times 1$ pair sits in the relaxed cell of the enumerated AFM configuration ($a = 8.372$, $c = 8.376$ Å) with a mesh equivalent to 6 points along $b$ in the primitive cell, the $1 \times 4 \times 1$ pairs in the relaxed FM cell ($a = 8.407$, $c = 8.396$ Å) with 8. The same-cell $+-+-$ check is compared with the FM member of the $++--$ pair; its relaxation, first cut off at ionic step 24, was continued to the force criterion (28 ionic steps, final maximum force 0.007 eV Å$^{-1}$), lowering $\Delta E$ by

0.48 meV from the cut-off value.

On the loosely relaxed (ISIF= 3, ISMEAR= −5) $Pm$ geometry used for the spiral probes, whose $Pm$ mirror is normal to $b$ and contains the Fe site, Moriya's rules allow any $\mathbf{D} \perp b$, and two projections determine $|\mathbf{D}|$. Each was obtained from a pair of 90° spin spirals in the $1 \times 4 \times 1$ cell (four Fe, moments constrained with `I_CONSTRAINED_M`= 1, $\lambda = 10$, SOC on, `EDIFF`= $10^{-7}$ eV, single points on the relaxed FM structure), rotating in the plane normal to $\hat{n}_1 \parallel a$ or $\hat{n}_2 = \hat{b} \times \hat{n}_1$ with either chirality. For a 90° spiral $\hat{s}_i \times \hat{s}_j = \pm\hat{n}$ on every chain bond while the symmetric exchange (including $J_{2b}$, whose bonds are antiparallel in both chiralities) is identical for the two, so $\mathbf{D}\cdot\hat{n} = [E(+) - E(-)]/8$ per bond. The constraint penalty energies were $\leq 1.6 \times 10^{-9}$ eV, the Fe moments had no component off their target axes (to the printed $0.001\,\mu_B$) with $|m_{\mathrm{Fe}}| = 1.50 \pm 0.01\,\mu_B$ in all four calculations, and the induced Tb moments ($\leq 0.123\,\mu_B$; Bi $\leq 0.002\,\mu_B$) rotate with the spiral, their magnitudes agreeing between chiralities to $0.001\,\mu_B$. The two projections, $\mathbf{D}\cdot\hat{n}_1 = -0.074$ and $\mathbf{D}\cdot\hat{n}_2 = -0.063$ meV, give $|\mathbf{D}| = 0.098$ meV. A symmetry analysis of the relaxed structures shows that this value is not a property of the compound. The loose geometry departs from $P\bar{6}2m$ by at most 0.029 Å (rms 0.014 Å) in the atomic positions and by 0.1% in the cell ($a = 8.4068$, $c = 8.3956$ Å, $\gamma = 119.96°$), whereas the tight (ISIF= 2) relaxation is $P\bar{6}2m$ to within 0.0065 Å, as is mp-22972 (rms 0.0034 Å). In $P\bar{6}2m$ the Fe chain lies on the threefold axis and every chain-bond midpoint on the $\sigma_h$ mirror, which is perpendicular to the bond, so $\mathbf{D} = 0$; the probes measured the DMI that the relaxation residual induces. The spin-dynamics sweep, which included this $\mathbf{D}$, is unaffected (Results).

Both $Tb_6FeBi_2$ entries, mp-22972 (stored $P\bar{6}2m$, FiM label) and mp-1079285 (stored $Pm$, FM label), relax to the same structure (0.0085 Å rms), but their tight pairs disagreed: in mp-22972 the AFM stripe that makes four of six inter-chain bonds antiparallel lay 0.46 meV atom$^{-1}$ below FM in an 18-atom orthorhombic cell with a $6 \times 2 \times 1$ mesh (a single $k$-point along the 14.5 Å axis), while the same stripe in the $2a$ cell of mp-1079285 lay 1.25 meV atom$^{-1}$ above FM ($2 \times 6 \times 3$). Because the compound is metallic, we recomputed both pairs as static single points on the tight relaxed structures at a uniform $k$-length of ∼60 Å along every axis ($15 \times 8 \times 5$ and $4 \times 15 \times 8$; same functional, $U$ and cutoff, EDIFF= $10^{-6}$ eV). The resulting energies are −94.44827 (FM) and −94.42743 eV (AFM) per cell for mp-22972 and −94.45633 and −94.43267 eV for mp-1079285, with Fe moments of 1.57–1.69 $\mu_B$ in all four and induced Tb moments ($\approx 0.05\,\mu_B$ in mp-22972)

antiparallel to Fe. Both entries are therefore FM, by 1.16 and 1.31 meV atom$^{-1}$; the 13% difference between them is the residual cell and mesh dependence. Two further AFM orderings of mp-22972, from its MP-recipe enumeration, lie 2.94 and 4.4 meV atom$^{-1}$ above FM. The mp-22972 AFM static was run in restarted one-hour segments; its moments were checked in every segment, and its FM–AFM difference is corroborated by the uninterrupted mp-1079285 pair. At the converged mesh the mp-1079285 margin implies $J_\perp \approx 2.94$ meV, 5% above the production value. The other metallic tight pair, mp-682554 ($3 \times 1 \times 2$ mesh, $k$-length 20–28 Å), was recomputed in the same way at $9 \times 3 \times 6$ ($k$-length 61–84 Å; static single points on the tight relaxed structures, EDIFF$= 10^{-6}$ eV, each started from the charge density of the production run). The energies are $-930.50887$ (FM) and $-930.55002$ eV (AFM) per 130-atom cell, with Fe moments of 3.07 and $\pm 3.04\,\mu_B$ and total moments of 8.00 and $0.00\,\mu_B$; the margin moves from $-0.313$ to $-0.317$ meV atom$^{-1}$ (41.1 meV per cell), so this AFM verdict is $k$-converged.

**Exchange of mp-850225 and the unresolved spiral**

$FeP_6(WO_8)_3$ has one Fe per rhombohedral cell, so its Fe sublattice is a Bravais lattice on which a two-sublattice canting probe sees no DMI (the $+\mathbf{t}$ and $-\mathbf{t}$ bonds contribute opposite $\mathbf{D} \cdot (\mathbf{S}_0 \times \mathbf{S}_1)$). The two exchange shells are separated by two same-cell pairs in 68-atom supercells: the first, from the ordering enumeration, gives $\Delta E = 8(J_1 + J_2)$; the second, with alternate hexagonal Fe layers antiparallel along $c$, every out-of-plane bond and no in-plane one ($\Delta E = 12J_2$). Resolving $D$ needs a spin spiral in the 102-atom hexagonal cell. That spiral pair (noncollinear with SOC, Fe directions constrained, W moments free) was run in three attempts (about 65 node-hours): one chirality converged (EDIFF $= 10^{-6}$ eV), but the other never reached self-consistency at the precision the extraction needs ($\Delta E = 18 \sin 120^\circ D_z \approx 15.6\,D_z$, i.e. about 1.6 meV for $D \sim 0.1$ meV), so no DMI is quoted for this compound.

**Finite-temperature spin dynamics**

The DFT-derived $J = 2.78$ meV, with no fitting or rescaling, parameterizes a twelve-neighbour Heisenberg Hamiltonian for the Fe sublattice of mp-754090 (frame-

work Li/Te/W/O sites carry no spin and are omitted from the geometry), implemented in LAMMPS-SPIN [33] within LAMMPS [72, 73] via the `spin/exchange` pair style (which counts each bond once, matching the convention of the DFT extraction) with a 5.5 Å cutoff that captures all twelve Fe–Fe bonds per site. `spin/exchange` uses the Bethe–Slater radial form $J(r) = 4a\,(r/d)^2 e^{-(r/d)^2}$ (with its $b$ parameter set to zero); $d = 5.274$ Å is solved so that $J(5.130\ \text{Å}) = J(5.418\ \text{Å})$ exactly and $a$ set so that both equal the DFT $J$, which makes $J(r)$ constant to within 0.06% over the two shells. A zero-step energy evaluation of the FM and Fe0-reversed configurations of the model reproduces the DFT $E_{\text{AFM}} - E_{\text{FM}} = 16J = 44.54$ meV per cell to 0.003%. The DMI is not included in the production sweep: the LAMMPS `spin/dmi` pair style implements $\mathbf{D}_{ij} = D\,(\hat{e}_{ij} \times \hat{v})$ with a single global vector $\hat{v}$, and for the DFT probe geometry ($\hat{s}_i \times \hat{s}_j = \pm\hat{y}$) its energy difference is $D\,(\hat{v} \times \hat{y}) \cdot \sum_b \hat{e}_b$ over the eight inter-sublattice bond directions – a sum that is 0.03 (in units of a bond vector) for this near-centrosymmetric neighbour shell, so no choice of $\hat{v}$ can reproduce the DFT $E(-90°) - E(+90°)$ (a zero-step evaluation with $D = 0.097$ meV and $\hat{v} = \hat{y}$ gives 0.000 meV); a faithful model needs the eight bond-resolved DMI vectors. We instead bound the effect of a DMI of the measured magnitude on $|m|(T)$ with a control sweep that adds `spin/dmi` with $D = 0.097$ meV and $\hat{v} = \hat{y}$ (an arbitrary but representative placement of a 3.5% antisymmetric term, the $y$-projection of the first probe) at four temperatures. The model is run on a $16 \times 8 \times 8$ replication (2048 spins) of the two-atom Fe primitive cell, with periodic boundaries, a rigid (spin-only) lattice and zero single-ion anisotropy. Each run starts from the fully aligned FM ground state and is evolved under Langevin spin dynamics ($dt = 0.1$ fs) – stochastic Landau–Lifshitz–Gilbert dynamics of classical spins [74, 75], the standard atomistic approach to finite-temperature magnetism [76, 77] – for 200,000 steps (20 ps) at twelve temperatures from 10 to 400 K; the normalized net magnetization $|m| = |\sum_i \hat{s}_i|/N$ is averaged over the second half of each run and over three independent thermostat seeds per temperature. The Langevin damping is $\alpha = 1$: the equilibrium $|m|(T)$ does not depend on $\alpha$, but the thermalization time of the softest magnons of the finite cell scales as $1/\alpha$ (see Spin-model validation checks). The independent Metropolis Monte Carlo runs 2000 equilibration and 2000 measurement sweeps per temperature on the same 2048-spin lattice.

For mp-1079285 the same machinery carries a three-shell model,

$$H = -\Big(J_b \sum_{\text{chain}} + J_{2b} \sum_{\pm 2b} + J_\perp \sum_{\text{inter}}\Big)\hat{s}_i \cdot \hat{s}_j + \sum_{\text{chain}} \mathbf{D}\cdot(\hat{s}_i \times \hat{s}_j), \tag{3}$$

on the Fe sublattice (one spin per primitive cell, $|m| = 1.68\,\mu_B$ as a classical unit vector). Two `spin/exchange` instances carry the chain bond (cutoff 5 Å) and the 8.4 Å shell (Bethe–Slater radial form with its root placed at the chain distance); because the $\pm 2b$ bond has the same length as the inter-chain bonds, spins carry an atom type equal to their chain index modulo four, so that $\pm 2b$ pairs (types differing by two) receive $J_{2b}$ by an explicit per-type override while inter-chain pairs (equal types) receive $J_\perp$. The DMI is carried by `spin/dmi` on the chain bond only (5 Å cutoff), with its vector calibrated – not assumed – by zero-step energy evaluations of the very 90° spirals used in the DFT, so that the model reproduces $E(+) - E(-)$ for both spiral planes. Before any sweep, zero-step evaluations of the $1 \times 2 \times 1$, $1 \times 4 \times 1$ $(++--)$, $2a$ and $2c$ cells reproduce the four DFT energy differences – the two chain cells exactly, the $2a$ and $2c$ cells to 0.5%, which is the residual of fitting one $J_\perp$ to two inequivalent shells. The production cell is an $8 \times 8 \times 8$ replication of the $1 \times 4 \times 1$ unit (2048 spins), started from FM along $b$, with the same Langevin protocol at fourteen temperatures from 5 to 300 K. The same-cell parametrization was swept at 10, 100, 125, 150 and 200 K with three seeds each.

**Finite-size scaling**

Sweeps were repeated at $N = 256$, 864, 2048 and 4000 spins with the box aspect ratio held fixed, so that every size sees the same neighbour shells, as 40 ps Langevin runs ($\alpha = 1$) with the moments pooled over the last 20 ps of four thermostat seeds; for mp-1079285 the 135–150 K window, where the crossings lie, was run with 10–14 seeds per point, six of them 80 ps. Error bars on $U_4$ are a drop-one-seed jackknife. $U_4$ of a three-component order parameter tends to 2/3 in the ordered phase and 4/9 in the disordered one and is size-independent at $T_c$. Because $U_4$ is a ratio of small fluctuating moments, its jackknife error grows from ±0.0003–0.002 at 95 K to ±0.03 above $T_c$ for mp-754090 (from ±0.0002–0.0007 at 120 K to ±0.004–0.028 at 150 K for mp-1079285), and in that noisy regime the difference between two sizes can change sign on scales far below its own error bar. The quoted crossings are therefore the lowest-temperature sign changes that persist: the difference is required to

stay positive at a majority of the higher temperatures measured, which discards both the sub-noise flips inside the ordered plateau, where every $U_4$ has converged on 2/3, and isolated single-point reversals above $T_c$; one such non-persistent sign change was discarded for mp-1079285, whose 155 and 160 K points retain only four seeds each and scatter back above 4/9. $T_c$ and its uncertainty are the mean and standard deviation of the three pairwise crossings of consecutive sizes. The longer runs at 135–150 K cut the $U_4$ error where the crossings lie ($\pm 0.048 \to \pm 0.014$ for $N = 4000$ at 145 K), though not uniformly: the same cell's error at 150 K grew from $\pm 0.020$ to $\pm 0.028$ as the longer runs sampled more of the disordered-phase excursions.

Re-deriving both crossings under changes to the sampling window (Supplementary Note 3) shows that the half-run equilibration cut is required and that the autocorrelation time of $U_4$ near $T_c$ is comparable to the run length, so the quoted spreads are lower bounds.

### Spin-model validation checks

Each model passes four checks before any sweep (Supplementary Note 4 describes the errors they caught): a zero-step energy evaluation of the model against every DFT pair it was built from (for mp-1079285 the $++--$ check is what exposes a $\pm 2b$ bond silently given $J_\perp$); an explicit count and geometry of the neighbour shells the cutoff admits; an independent Metropolis Monte Carlo calculation of $|m|(T)$ on the same lattice; and a window-drift test on every reported point, comparing the mean of $|m|$ over consecutive 2 ps windows of the averaged half. For the mp-1079285 production sweep the seed-averaged drift is $\leq 0.008$ at every temperature except $+0.018$ at 150 K, against within-run fluctuations of $\pm 0.05$ there, with no monotonic trend; for the same-cell sweep it is $\leq 0.008$ except $+0.05$ at 150 K, the transition point, against a within-run fluctuation of $\pm 0.04$; for mp-754090 the 100 K point has a within-run fluctuation of 0.043 and window means that differ by up to 0.049.

## DATA AVAILABILITY

The per-candidate DFT verdict tables, the spin-model parameter files and the reduced magnetization and finite-size-scaling sweeps behind every figure are available at https://github.com/akhtar07/magnetic-materials-dmi-screening and are provided as Source

Data with this paper. The harmonized database is rebuilt from the seven public sources by the code below.

## CODE AVAILABILITY

The database harmonization, classifier (LightGBM and graph-neural-network), screening, VASP-input generation, spin-model, spin-dynamics, finite-size-scaling and figure scripts are available at https://github.com/akhtar07/magnetic-materials-dmi-screening.


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## AUTHOR CONTRIBUTIONS

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## COMPETING INTERESTS

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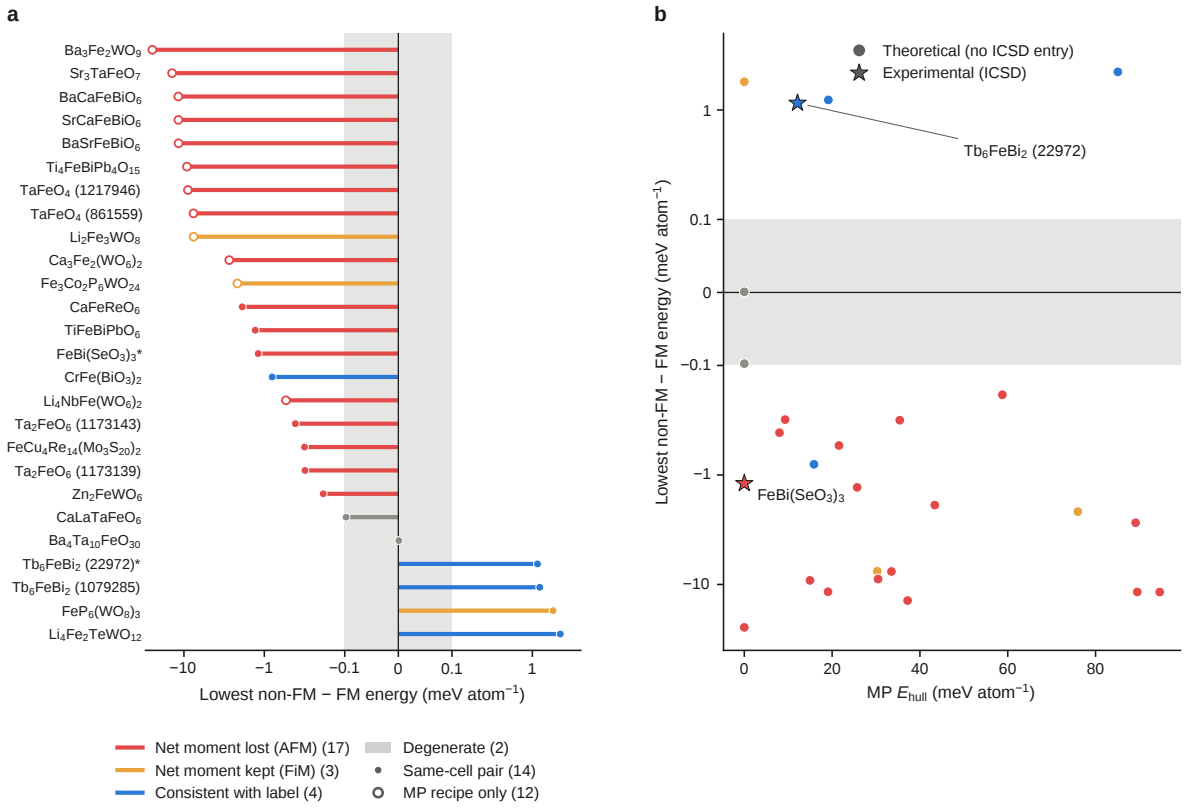



Fig. 2. **DFT verdicts on the ordering labels of 26 Materials Project candidates. a**, Energy of the lowest audited non-FM configuration relative to FM, per atom, for each candidate (symmetric-logarithmic axis, linear within $\pm 0.1$ meV atom$^{-1}$); negative values place a non-FM collinear state below FM. Colours give the outcome class of Table 1. Ferrimagnetic ground states are orange, or blue for the FiM-labelled mp-551086 and mp-22972; the orange bar above zero is mp-850225, whose Fe sublattice is FM while its W moments are antiparallel to the Fe. Only calculations that converged and passed the spin-state audit enter, and a same-cell pair replaces the MP-recipe number wherever one exists. The gray band is the $\pm 0.10$ meV atom$^{-1}$ degeneracy window; asterisks mark the two experimentally reported (ICSD) compounds; duplicate formulas carry their MP identifier. The two $Tb_6FeBi_2$ entries, which relax to the same structure, agree at a converged $k$-mesh (Methods). Filled end markers, margins from a same-cell (tight) pair; open, MP-recipe comparison only. **b**, The same margin against MP's stored hull distance. Stars, ICSD compounds; circles, hypothetical structures with no ICSD entry. The margin shows no trend with hull distance. $n = 26$ candidates; one point per candidate.

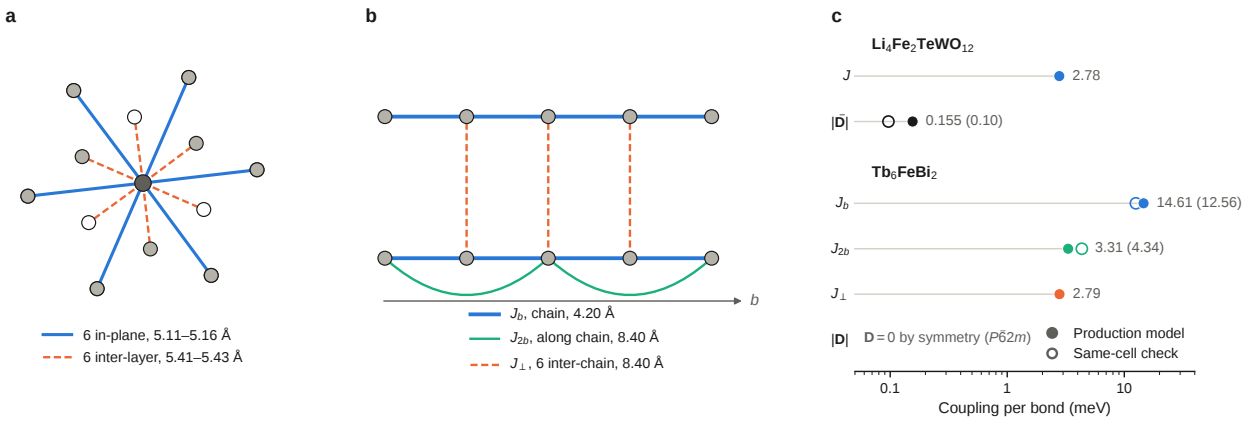



Fig. 3. **Exchange paths and DFT couplings of the two modelled ferromagnets. a**, $Li_4Fe_2TeWO_{12}$ (mp-754090): the twelve Fe neighbours of one Fe (centre), projected on the plane of the six in-plane bonds (solid); dashed, the six inter-layer bonds; filled and open circles lie above and below that plane. One same-cell FM/AFM pair fixes a single $J$ for all twelve bonds; six pairs of constrained canting probes resolve the bond-averaged DMI vector $\bar{\mathbf{D}}$. **b**, $Tb_6FeBi_2$ (mp-1079285): two neighbouring Fe chains along $b$, drawn to scale, with the chain bond $J_b$, the next-nearest chain bond $J_{2b}$ (arcs, shown on one chain) and the inter-chain bond $J_\perp$ (dashed; six per Fe, one direction drawn); in $P\bar{6}2m$ the chain-bond DMI vanishes by symmetry. **c**, The couplings per bond on a logarithmic axis. Filled markers, the production spin models; open markers, the $J_b$ of the same-cell $+-+-$ check and the $J_{2b}$ it implies, which bracket the $Tb_6FeBi_2$ model, and for $Li_4Fe_2TeWO_{12}$ the $y$ projection $D_y$ from the first canting pair alone.

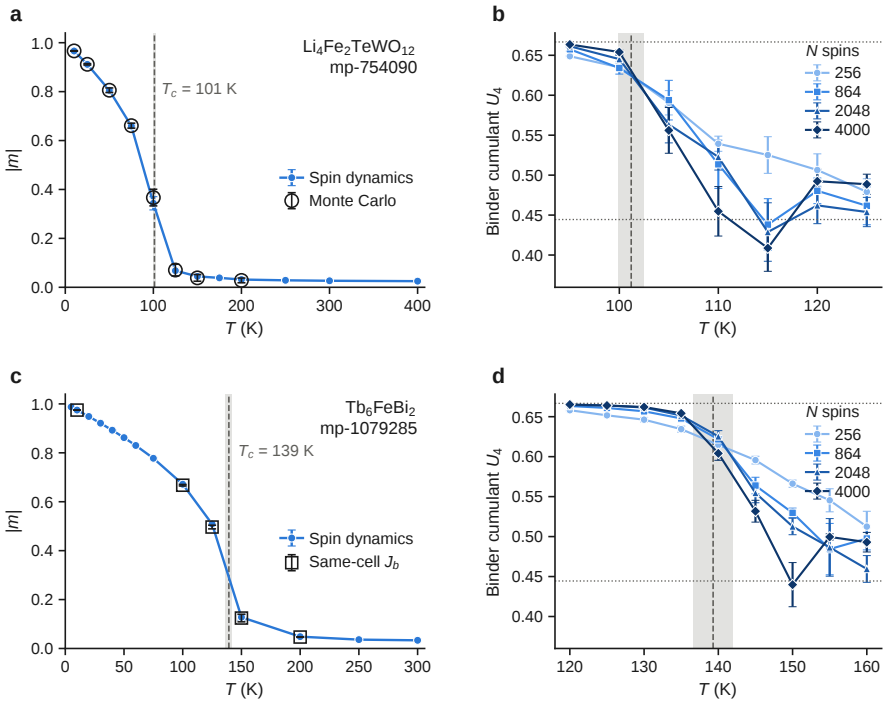



Fig. 4. **Finite-temperature ordering of the two DFT-parameterized spin models. a**, $Li_4Fe_2TeWO_{12}$ (mp-754090): normalized Fe-sublattice magnetization $|m|$ of the 2048-spin cell from 20 ps Langevin dynamics ($\alpha = 1$) started from the ordered state, averaged over the last 10 ps; filled circles, mean ± s.d. over $n = 3$ thermostat seeds per temperature (error bars mostly smaller than the symbols); open circles, independent Metropolis Monte Carlo of the same Hamiltonian on the same lattice (mean ± s.d. over 20 blocks of the 2000 measurement sweeps). **b**, Binder cumulant $U_4$ at four system sizes ($N$, box aspect ratio fixed) from 40 ps runs, moments pooled over the last 20 ps of $n = 4$ seeds; error bars, drop-one-seed jackknife. Dotted lines, the ordered (2/3) and disordered (4/9) limits of a three-component order parameter. **c**, **d**, As **a**, **b** for the three-shell $Tb_6FeBi_2$ (mp-1079285) model; open squares in **c**, the same-cell parametrization ($n = 3$ seeds). In **d**, $n = 10$–14 seeds at 135–150 K (six of them 80 ps) and $n = 4$ elsewhere. In every panel the dashed line and gray band are $T_c$ and its spread, the mean ± s.d. of the three pairwise crossings of consecutive sizes.

## SUPPLEMENTARY INFORMATION

### Supplementary Note 1: MAGNDATA retrieval

A new end-to-end scraper against the live Bilbao Crystallographic Server index recovered 1,919 of 2,414 entries (79.5%). The 495 failures break down as 449 HTTP 404s from the server's temporary `.mcif` endpoint, 41 structures the parser refuses (36 explicitly disordered, 5 incommensurate), 3 read timeouts, and 2 entries whose fetched CIF contained no structure. Within the 404s, 148 are entries whose formula carries fractional site occupancies and 3 carry HTML markup in the upstream formula string, which makes the request URL malformed; those 151, together with the 41 refusals, are the at most 192 entries attributable to a data-format limitation. The remaining 298 are 404s on well-formed URLs for integer-stoichiometry entries, and they are transient rather than permanent: they persist across the scraper's four in-run retries, but re-requesting one of them (entry 0.57, $ScFeO_3$) returned 404 and then 200 within minutes, while its entry page is served throughout. 79.5% is therefore a property of this run rather than a ceiling. Nothing below depends on the missing entries, all of which are simply absent from the training and validation sets, but the MAGNDATA ground truth used here is smaller than the source allows.

### Supplementary Note 2: the JARVIS-DFT `ehull` field

The `ehull` field of the JARVIS-DFT snapshot used here is not a hull distance (its median over the 93,902-entry snapshot is 1.7 eV atom$^{-1}$, the known stable compound TbMnSi carries `ehull`= 2.45 eV atom$^{-1}$, and 101 of the 3,206 harmonized JARVIS records with a *positive* formation energy carry `ehull`= 0, among them $BaFeHg_4$ at +0.36 eV atom$^{-1}$); it was therefore discarded and the hull distance of every JARVIS record recomputed from JARVIS's own OptB88vdW formation energies with a per-chemical-system convex hull (13,967 of 19,116 records within 100 meV atom$^{-1}$). Three of the four JARVIS entries in the original top-30 shortlist fail this recomputed criterion ($Mg_6FeBi$ 0.13, $BaFeHg_4$ 0.36, $Ba_4ScHg$ 0.54 eV atom$^{-1}$); none of them entered the DFT batch, which is Materials Project only.

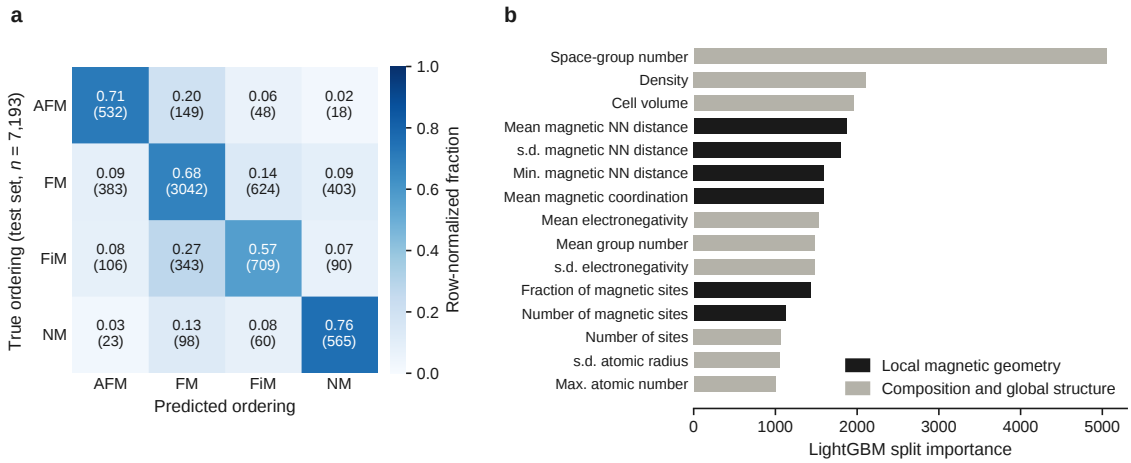
**Supplementary Note 3: sampling-window robustness of the Binder crossings**

Both crossings were re-derived under changes to the sampling window to test how much of the quoted spread is statistical. The equilibration cut matters and cannot be relaxed: discarding only the first quarter of each run instead of the first half moves mp-754090 to 109.8 K and mp-1079285 to 134.5 K, because the retained data still contain the thermalization transient from the ordered start. With the half-discard in place, mp-754090 is insensitive to further truncation – averaging only the last quarter gives 98.0/102.4/103.0 K, mean 101.1 K, against the published 101.2 K. mp-1079285 is not: its last quarter gives 141.0/141.1/123.4 K, mean 135.2 K, and the entire shift comes from the $2048 \times 4000$ pair. A check that shortens no window – analyzing only the six 80 ps runs, which cover 135–150 K – returns 140.0/143.2/135.7 K, mean 139.6 K, within 0.3 K of the published value. The instability of individual pairs is visible directly in the raw series: one 40 ps seed at $N = 4000$, 140 K has $U_4 = 0.582$ in its third quarter and 0.383 in its fourth (a 10 ps window each), so the autocorrelation time of $U_4$ near $T_c$ is comparable to the run length. Pooling 10–14 seeds suppresses the residual quarter-to-quarter difference of the pooled estimate to $\leq 0.039$ but does not eliminate it.

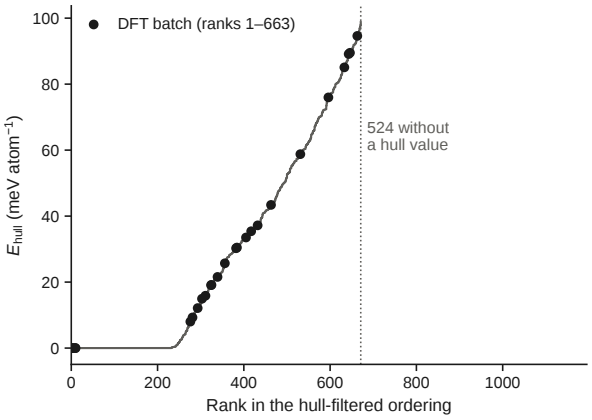
**Supplementary Note 4: errors caught by the spin-model checks**

The mp-754090 spin model was corrected twice during this work, and the Langevin protocol once; each error was caught by a check that we now apply to every model before a sweep. A first extraction quoted $J = \Delta E/4$, double-counting the flipped bonds ($J = 12.06$ meV, which placed the $|m|(T)$ knee at 150–200 K). The next used a 5.25 Å neighbour cutoff that keeps only the six coplanar in-plane bonds and drops the six inter-layer ones, so that $\Delta E = 8J$ and $J = 5.57$ meV – the correct sum of an in-plane and an inter-layer bond, but on a lattice of uncoupled two-dimensional triangular layers, which has no long-range order at any finite temperature; once properly equilibrated it placed the knee at 25–50 K, and the Monte Carlo on that lattice gives $|m|(10\text{ K}) = 0.64$ and 0.23 at 25 K, the no-long-range-order signature that exposed the cutoff error. Finally, at the conventional damping $\alpha = 0.05$ neither 4 ps nor 20 ps runs had equilibrated at the temperatures nearest the transition (at 100 K the seed-averaged $|m|$ was still falling across the averaged half of both:

from 0.49 in its first 0.5 ps to 0.38 in its last for the 4 ps run, and from 0.15 in its first 2 ps window to 0.07 in its last for the 20 ps run); those runs are kept in the repository and approach the reported values from above.




Supplementary Fig. 1. **Structure-aware ordering classifier on the held-out test set.** $n$ = 7,193 records from a stratified 80/20 split. **a**, Row-normalized confusion matrix (counts in parentheses). The dominant confusion is FM ↔ FiM, between the two net-moment classes that the screening funnel treats identically, whereas AFM and NM are recovered at 71% and 76%. **b**, The fifteen most-used features by LightGBM split count; black, the local-magnetic-geometry features, computed from atomic positions and species only; gray, composition and global-structure features.




Supplementary Fig. 2. **The DFT batch is not the most stable end of the funnel.** Hull distance against rank for the 1,195 candidates that survive the hard hull filter, in an ordering that sorts records with a known hull distance first (671) and by hull distance; the 524 records from sources without a hull value follow the dotted line. Black circles, the 26 DFT candidates, which span ranks 1–663.

Supplementary Table 1. **Ordering classifiers compared on one split.** All models trained on the same records (28,769; the 28,673 with an ordered structure for the graph-based models, of which CGCNN holds out 10% for validation) and scored on the 7,161 test records with an ordered structure (%). CGCNN, mean ± s.d. over three seeds; the LightGBM models are deterministic. MAGN. bal., balanced accuracy on the 369 experimentally labelled test records (311 AFM, 39 FiM, 19 FM); this column varies by up to 4 points with the training set and does not rank the models. MP FM→non-FM, fraction of Materials Project FM test records predicted AFM or FiM by the held-out model, the base rate for the next column. The last two columns (Overt., Conf.) are from retraining with the 26 DFT candidates held out: how many of the 20 candidates with a non-FM DFT ground state are predicted AFM or FiM, and how many of the 4 whose Fe sublattice remains FM are predicted FM (CGCNN, range over seeds). The screening used LightGBM v2 (on all 7,193 test records 67.4% and 68.0%).

| | | Bal. | Recall | | | | MAGN. | MP FM | Overt. | Conf. |
|---|---|---|---|---|---|---|---|---|---|---|
| Model | Acc. | acc. | AFM | FM | FiM | NM | bal. | →non-FM | non-FM | FM |
| LightGBM v2 (screening) | 67.4 | 68.0 | 70.9 | 68.4 | 56.9 | 75.7 | 56.0 | 23.2 | 9/20 | 2/4 |
| LightGBM, MACE emb. | 69.6 | 68.3 | 56.5 | 71.4 | 63.9 | 81.5 | 53.3 | 22.0 | 12/20 | 2/4 |
| LightGBM, v2 + MACE | 74.6 | 71.7 | 65.9 | 78.4 | 63.0 | 79.5 | 55.1 | 15.8 | 6/20 | 2/4 |
| CGCNN | $63.3 \pm 3.2$ | $65.9 \pm 1.0$ | $63.6 \pm 6.3$ | $62.0 \pm 6.6$ | $58.0 \pm 2.3$ | $79.9 \pm 4.5$ | $49.1 \pm 6.4$ | $32.7 \pm 6.7$ | 9–12/20 | 0–1/4 |

Supplementary Table 2. **Small-margin non-FM verdicts re-decided with same-cell pairs.** Margin $E^{\min}_{\text{non-FM}} - E_{\text{FM}}$ (meV atom$^{-1}$) of the eight candidates whose MP-recipe non-FM verdict had $|\text{margin}| < 2$ meV atom$^{-1}$, before and after the tight same-cell pair, for the configuration in whose cell the pair was built. Every sign is kept at the production $k$-mesh. [a]The metallic $Tb_6FeBi_2$ entry mp-22972 reverses at a uniform ∼60 Å $k$-length (+1.16, FM; Methods), and is therefore classed consistent. [b]The metallic mp-682554 pair at a $9 \times 3 \times 6$ mesh ($k$-length 61–84 Å; Methods) keeps its sign. For $Li_4NbFe(WO_6)_2$ another, recipe-only configuration (−0.54) sets the final margin.

| Compound | MP id | MP recipe | Same cell |
|---|---|---|---|
| $Ta_2FeO_6$ | mp-1173139 | −0.31 | −0.31 |
| $Ta_2FeO_6$ | mp-1173143 | −0.41 | −0.41 |
| $CrFe(BiO_3)_2$ | mp-551086 | −0.85 | −0.80 |
| $FeBi(SeO_3)_3$ | mp-1197024 | −1.19 | −1.19 |
| $CaFeReO_6$ | mp-1343817 | −1.77 | −1.88 |
| $Tb_6FeBi_2$ | mp-22972 | −0.14 | −0.46 (+1.16)[a] |
| $FeCu_4Re_{14}(Mo_3S_{20})_2$ | mp-682554 | −0.29 | −0.31 (−0.32)[b] |
| $Li_4NbFe(WO_6)_2$ | mp-755203 | −0.73 | −0.47 |

Supplementary Table 3. **Magnetization of the $Li_4Fe_2TeWO_{12}$ (mp-754090) spin model.** Normalized net magnetization $|m|$ of the 2048-spin Fe sublattice: 20 ps Langevin dynamics ($\alpha = 1$) per point from the ordered state, averaged over the last 10 ps; mean ± s.d. over $n = 3$ thermostat seeds; DFT-derived twelve-neighbour $J = 2.78$ meV, exchange only.

| $T$ (K) | $\lvert m\rvert$ | $T$ (K) | $\lvert m\rvert$ |
|---|---|---|---|
| 10 | $0.966 \pm 0.000$ | 150 | $0.044 \pm 0.002$ |
| 25 | $0.912 \pm 0.001$ | 175 | $0.039 \pm 0.001$ |
| 50 | $0.805 \pm 0.002$ | 200 | $0.032 \pm 0.002$ |
| 75 | $0.659 \pm 0.004$ | 250 | $0.029 \pm 0.000$ |
| 100 | $0.343 \pm 0.027$ | 300 | $0.027 \pm 0.001$ |
| 125 | $0.068 \pm 0.001$ | 400 | $0.025 \pm 0.000$ |

Supplementary Table 4. **Magnetization of the $Tb_6FeBi_2$ (mp-1079285) spin model.** As Supplementary Table 3, for the three-shell model ($J_b = 14.61$, $J_{2b} = 3.31$, $J_\perp = 2.79$, and the loose-geometry $|\mathbf{D}| = 0.098$ meV, which vanishes in the symmetric structure; see Methods).

| $T$ (K) | $\lvert m\rvert$ | $T$ (K) | $\lvert m\rvert$ |
|---|---|---|---|
| 5 | $0.988 \pm 0.000$ | 60 | $0.830 \pm 0.003$ |
| 10 | $0.975 \pm 0.000$ | 75 | $0.778 \pm 0.003$ |
| 20 | $0.949 \pm 0.001$ | 100 | $0.672 \pm 0.003$ |
| 30 | $0.921 \pm 0.001$ | 125 | $0.509 \pm 0.003$ |
| 40 | $0.892 \pm 0.002$ | 150 | $0.128 \pm 0.012$ |
| 50 | $0.862 \pm 0.002$ | 200 | $0.048 \pm 0.002$ |
| | | 250 | $0.036 \pm 0.001$ |
| | | 300 | $0.033 \pm 0.001$ |